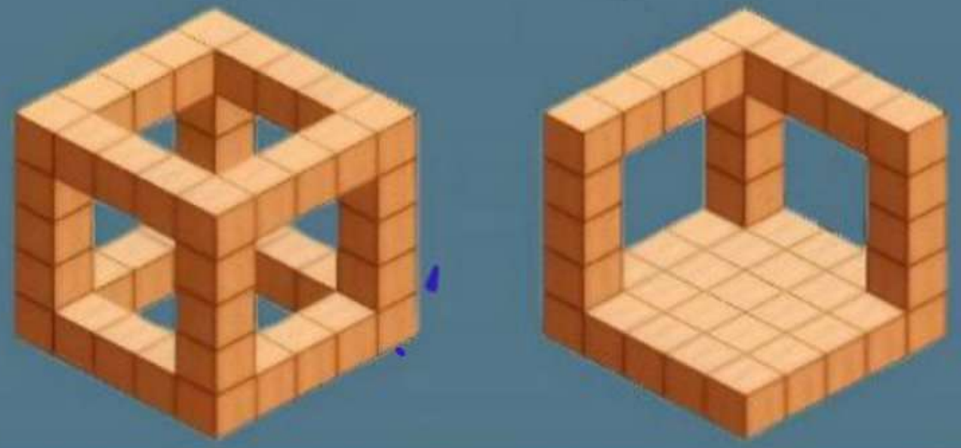
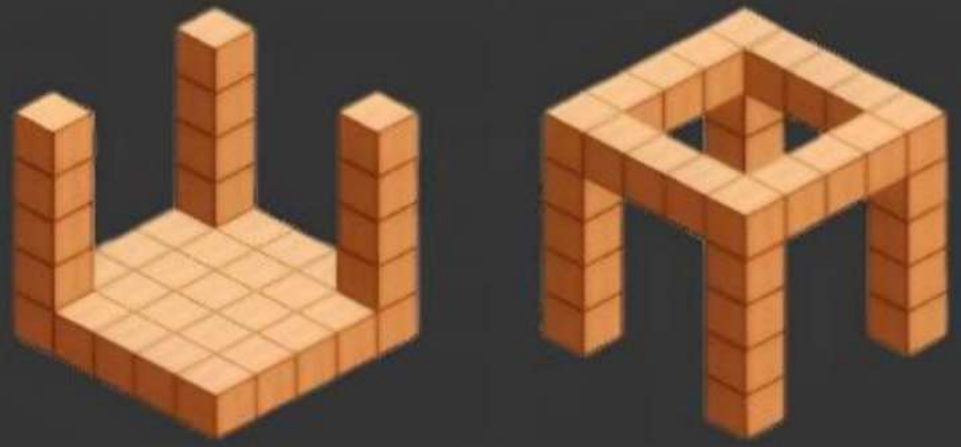
CAREERWILL
Easy to Learn...

COUNTING FIGURES





REASONING WITH PIYUSH VARSHNEY

SOLVE WITHOUT PEN & PAPER

Counting figures

Triangles (त्रिभुज)

↓

(3 Types)

★★★ # Triangles (त्रिभुज)

★★ # Squares (वर्ग)

★ # Rectangles (आयत)

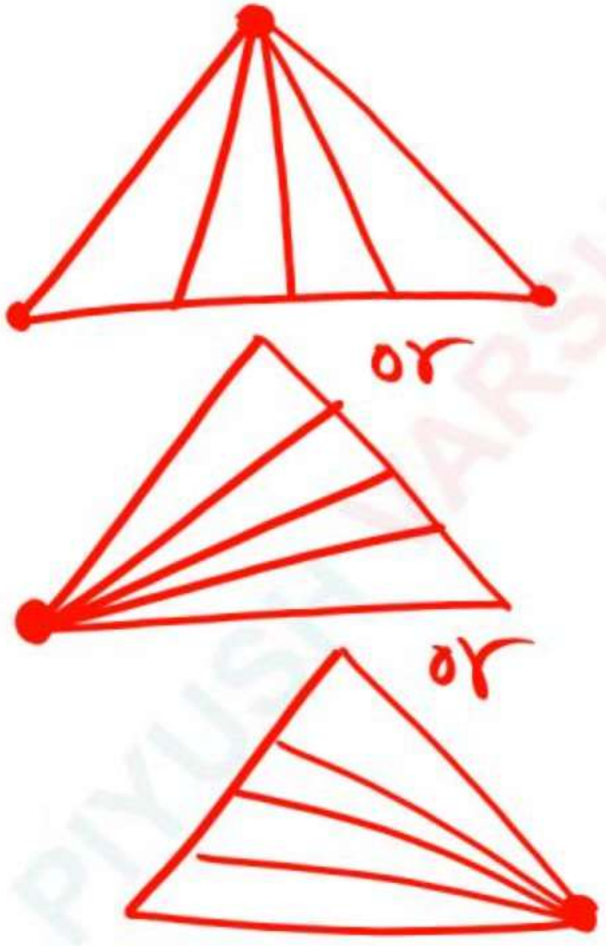
Straight lines

Circles

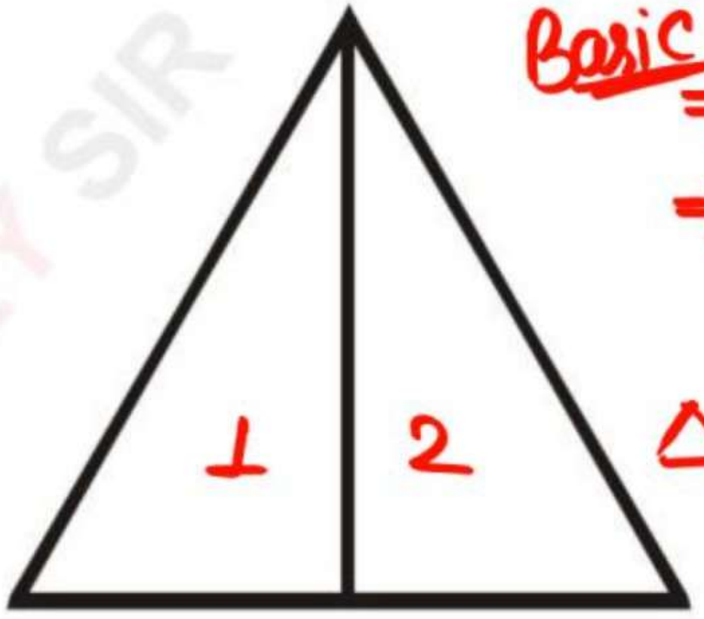
Blocks

Diagonals

Type-1



1.



Basic ⇒ 1, 2, 1+2

⇒ 3 त्रिभुज

△ = 1+2 ✓

= 3

Find out the number of triangles in the given figure?

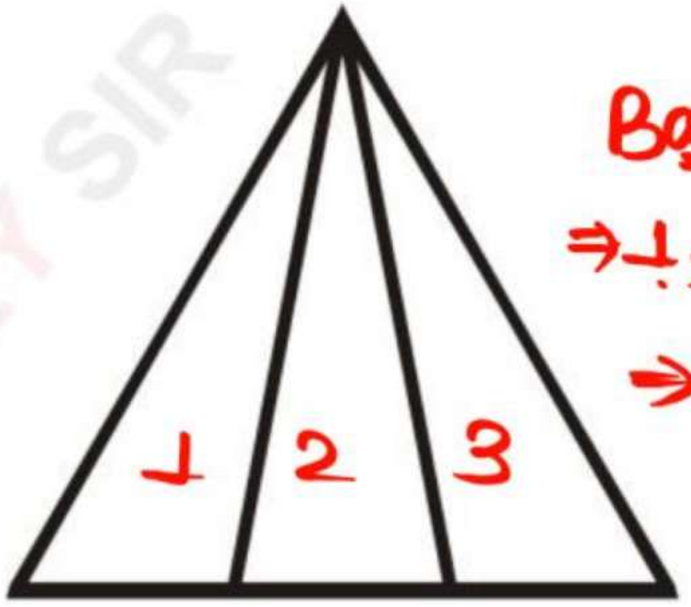
दी गई आकृति में त्रिभुजों की संख्या बताइए ?

(a) 2 (b) 4

(c) 3 (d) 5

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2.



Basic

⇒1, 2, 3, 12, 23, 123

⇒6

$\triangle = 1+2+3$

$= 6$

Find out the number of triangles in the given figure?

दी गई आकृति में त्रिभुजों की संख्या बताइए ?

(a) 5 **(b) 7**

(c) 6 **(d) 8**

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3.

$$\sum n = \frac{n(n+1)}{2}$$

$1+2+3+\dots+n = \frac{n(n+1)}{2}$

$\Delta = 1+2+3+4+5+6$

$= \frac{6\times 7}{2} = 21$

Basic

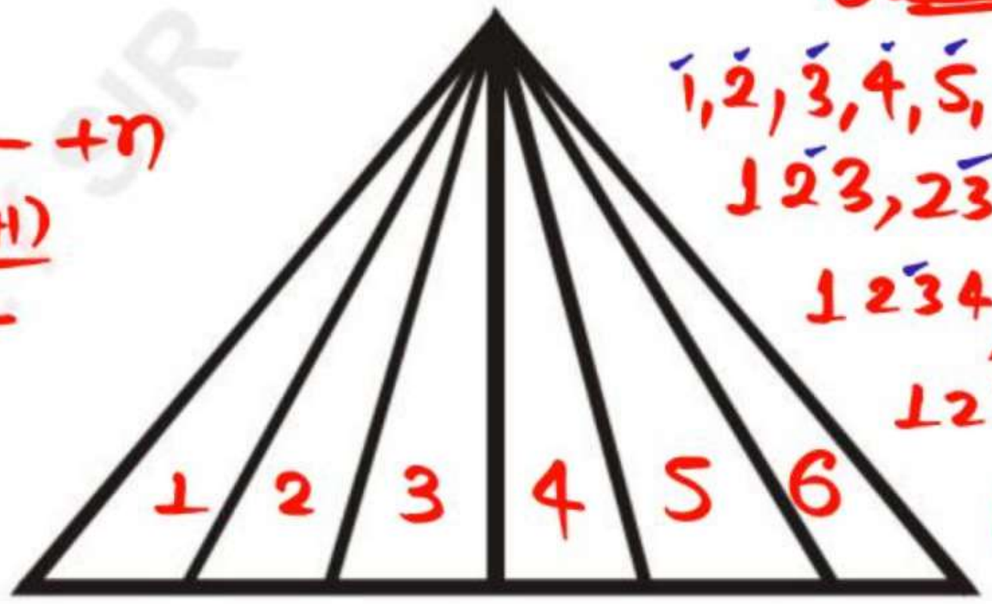
1, 2, 3, 4, 5, 6, 12, 23, 34, 45, 56

123, 234, 345, 456,

1234, 2345, 3456,

12345, 23456,

123456



Find out the number of triangles in the given figure?

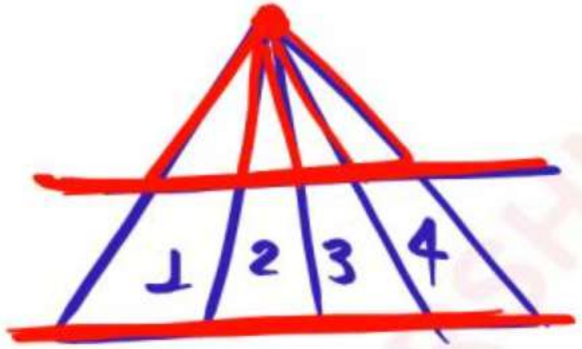
दी गयी आकृति में त्रिभुजों की संख्या बताइए?

(a) 23 **(b) 21**

(c) 18 **(d) 25**

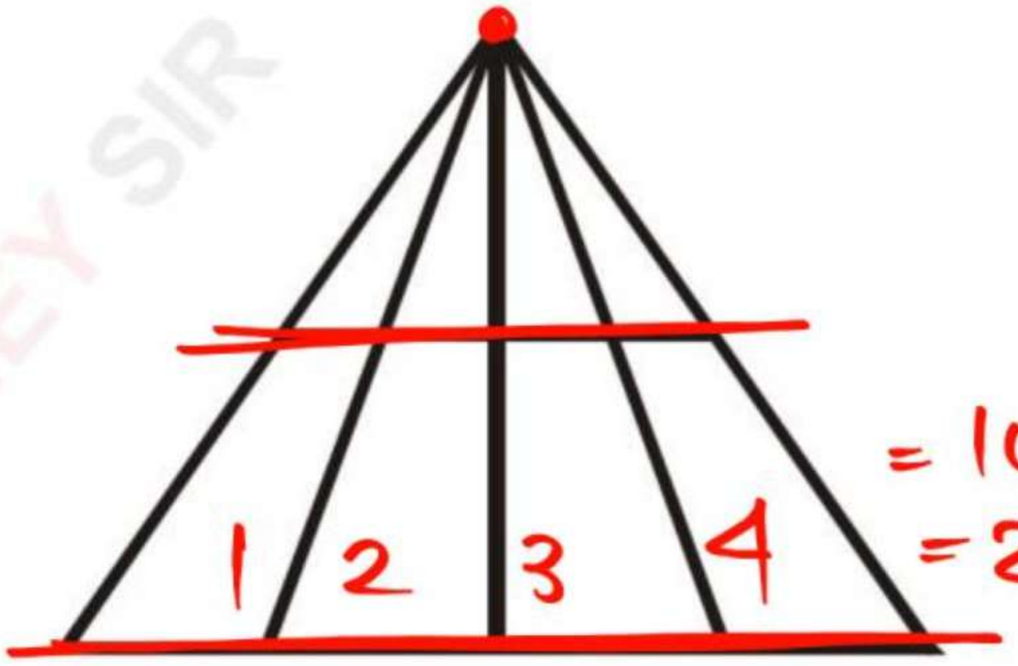
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4.



1+2+3+4

=10



=10X2

=20

Find out the number of triangles in the given figure?

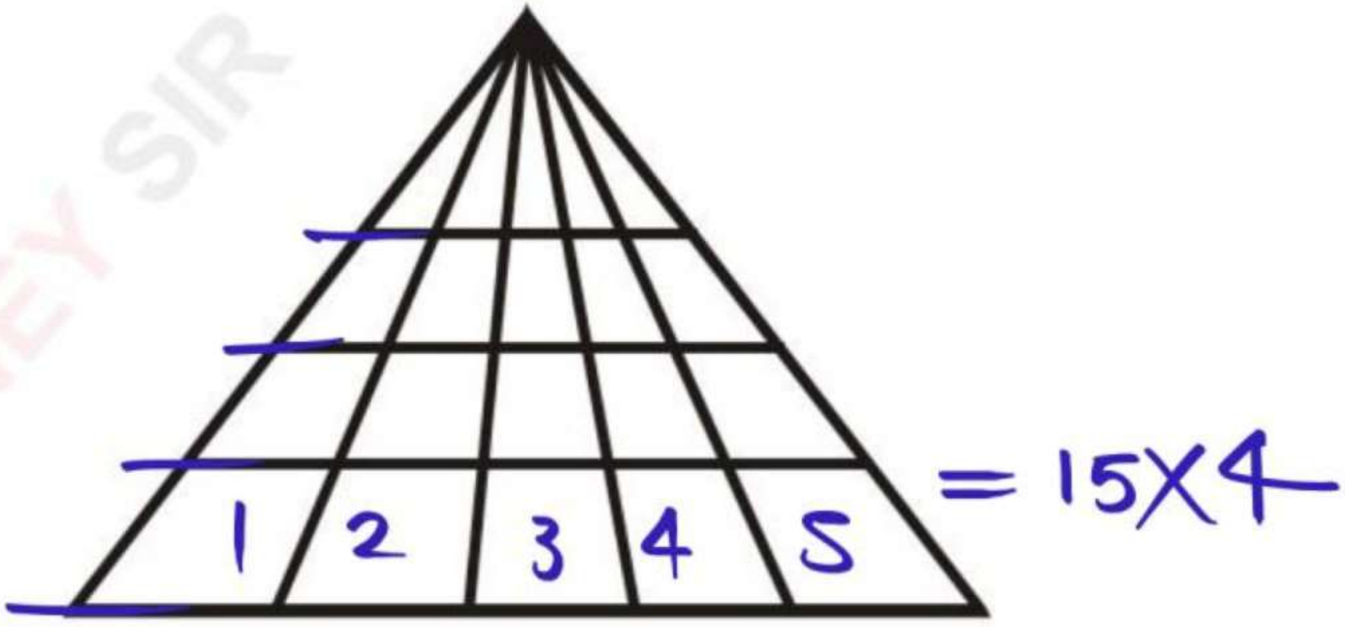
दी गयी आकृति में त्रिभुजों की संख्या बताइए?

(a) 18 **(b) 22**

(c) 20 **(d) 25**

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5.



Find out the number of triangles in the given figure?

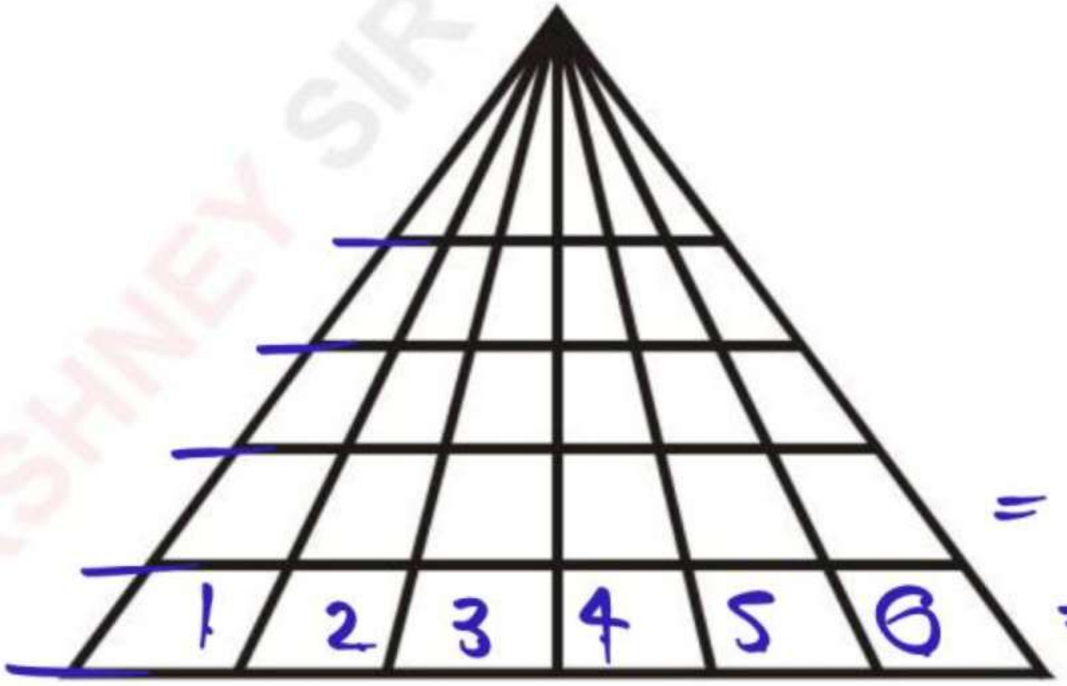
दी गयी आकृति में त्रिभुजों की संख्या बताइए?

(a) 60 **(b) 70**

(c) 50 **(d) 75**

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6.



[5-10 sec]

= 21 X 5

= 105

Find out the number of triangles in the given figure?

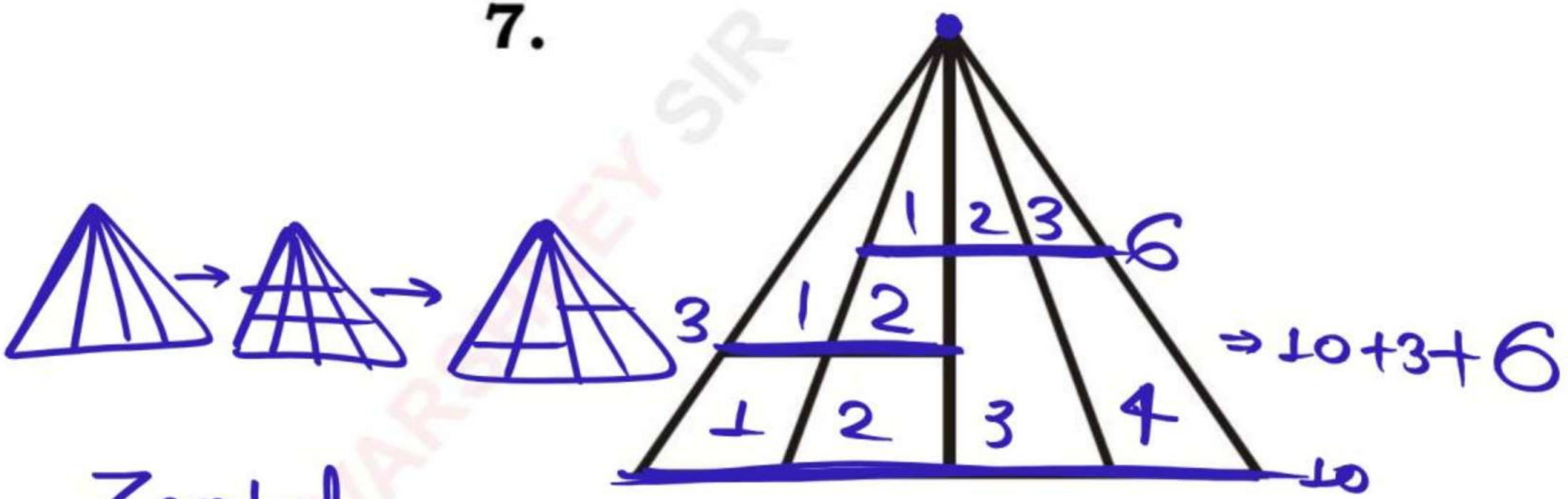
दी गई आकृति में त्रिभुजों की संख्या बताइए?

(a) 120 (b) 110

(c) 100 (d) 105

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7.



Zero level

Basic → Adv.

सारे Q. Exam के

Find out the number of triangles in the given figure?

दी गई आकृति में त्रिभुजों की संख्या बताइए?

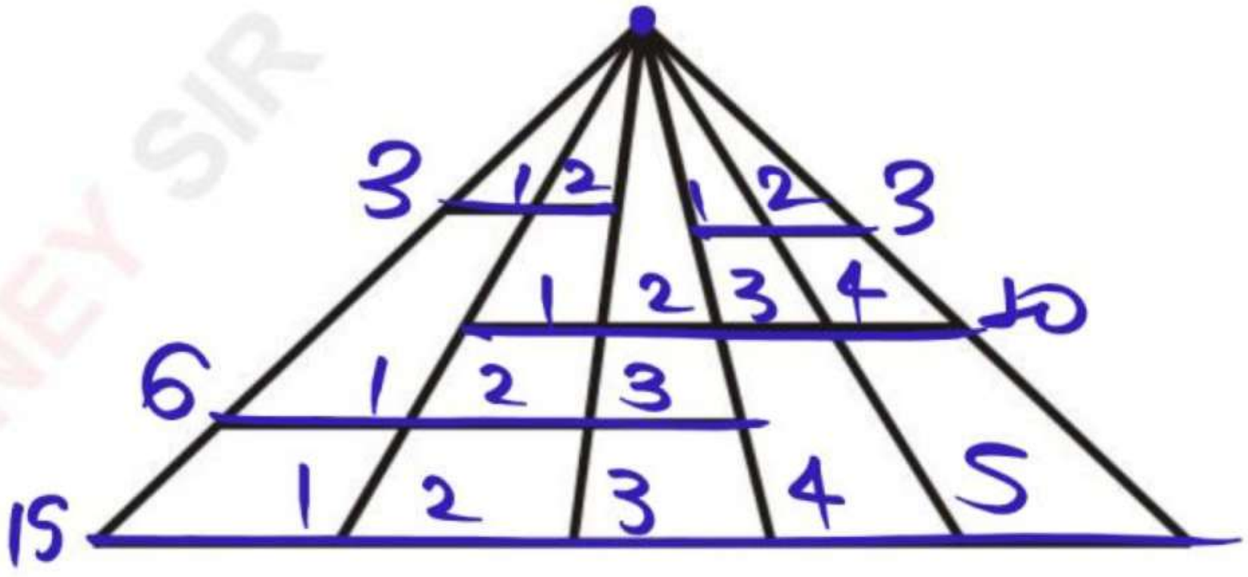
(a) 19 **(b) 20**

(c) 18 **(d) 21**

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8.



Find out the number of triangles in the given figure?

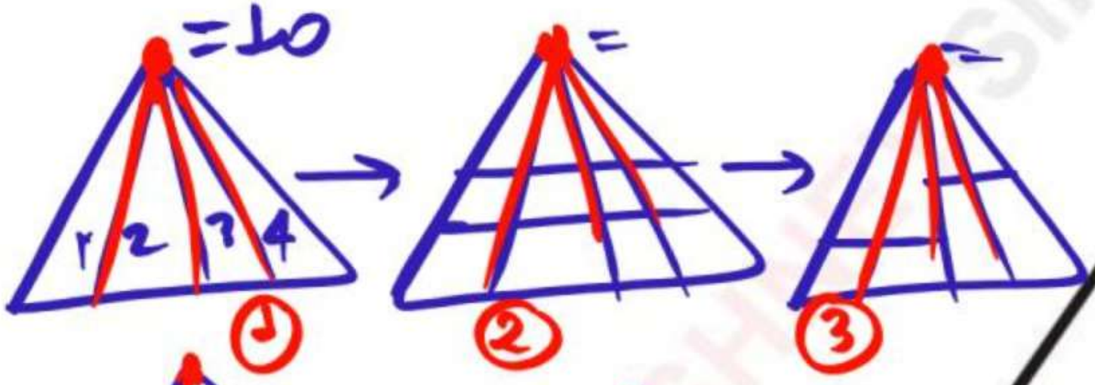
दी गई आकृति में त्रिभुजों की संख्या बताइए?

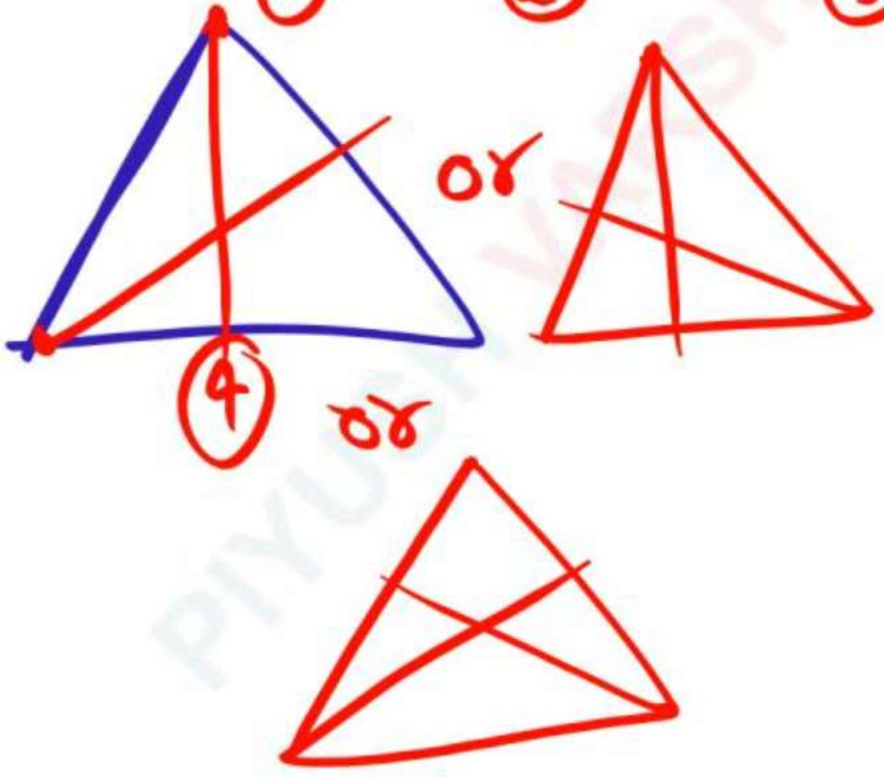
(a) 37 **(b) 40**

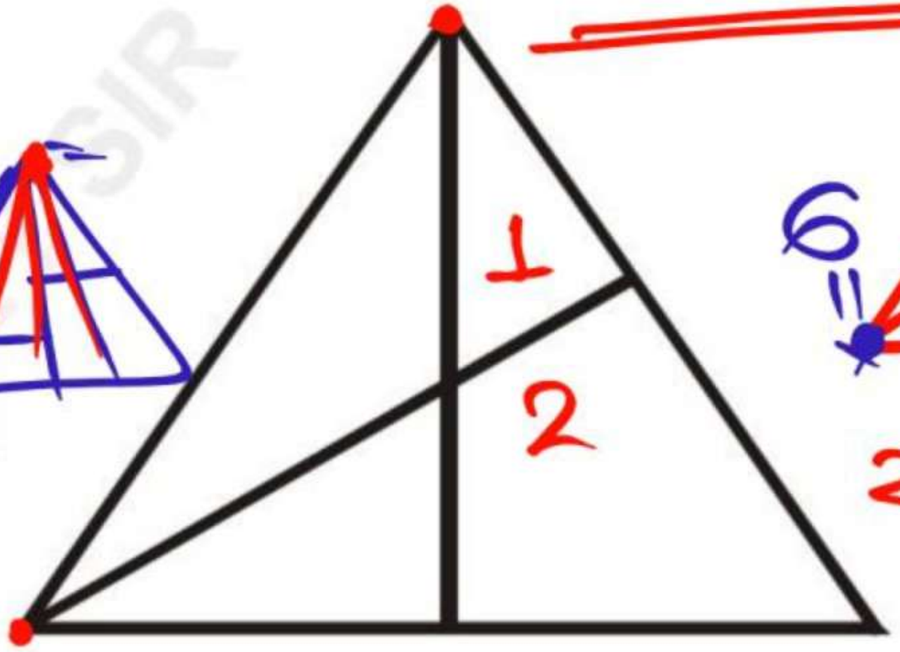
(c) 35 **(d) 42**

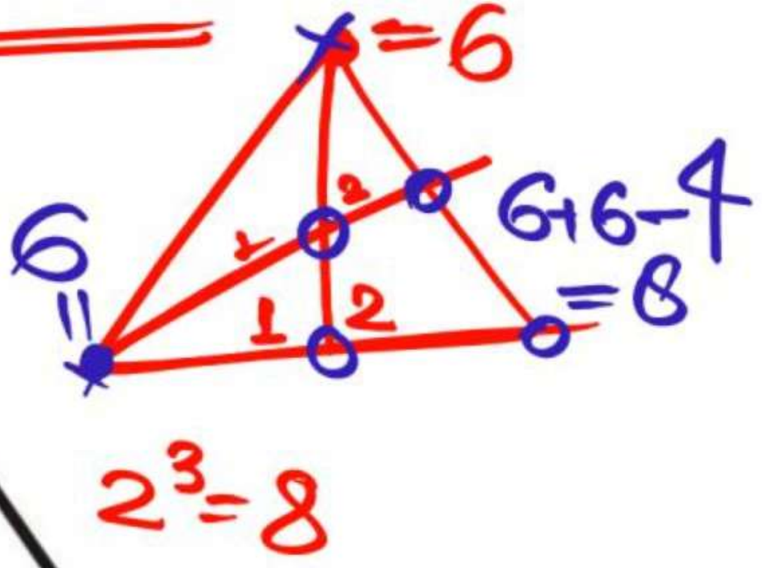
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9.









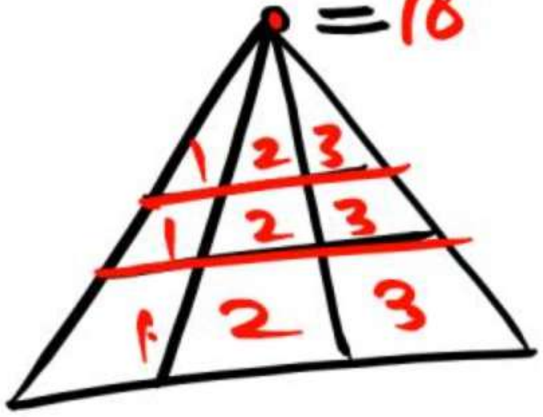
Find out the number of triangles in the given figure?

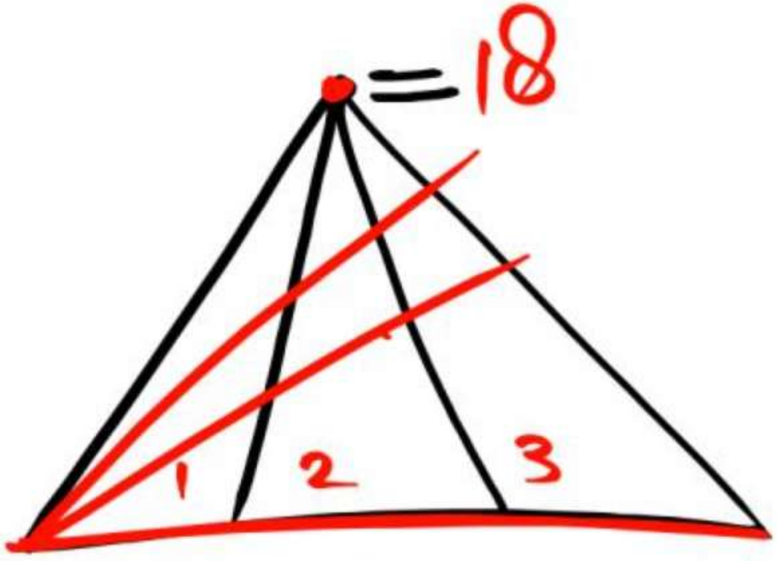
दी गई आकृति में त्रिभुजों की संख्या बताइए?

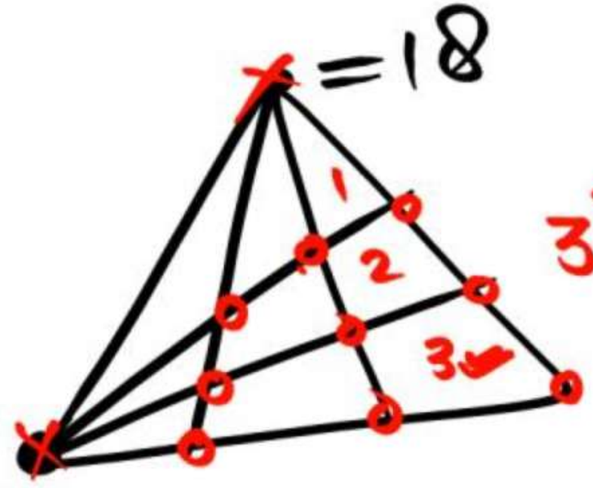
(a) 7 **(b) 8**

(c) 9 **(d) 6**

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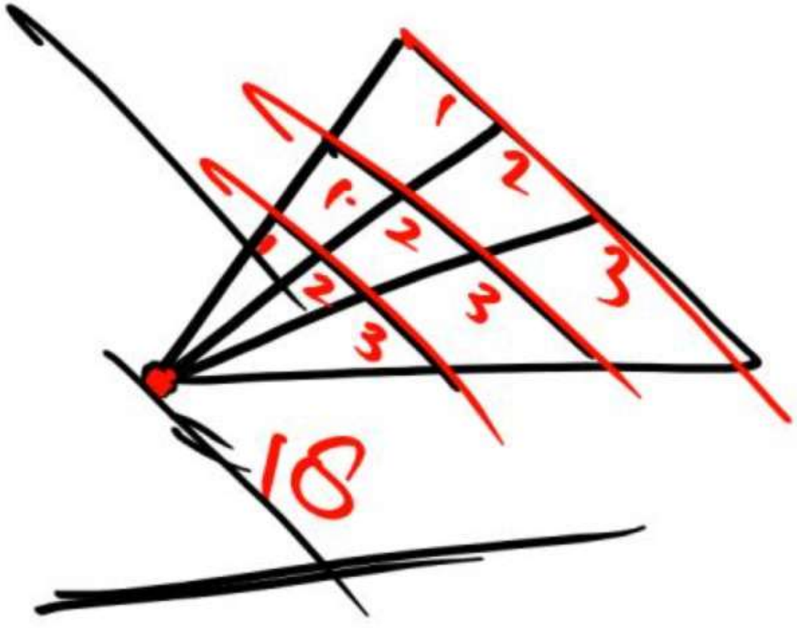


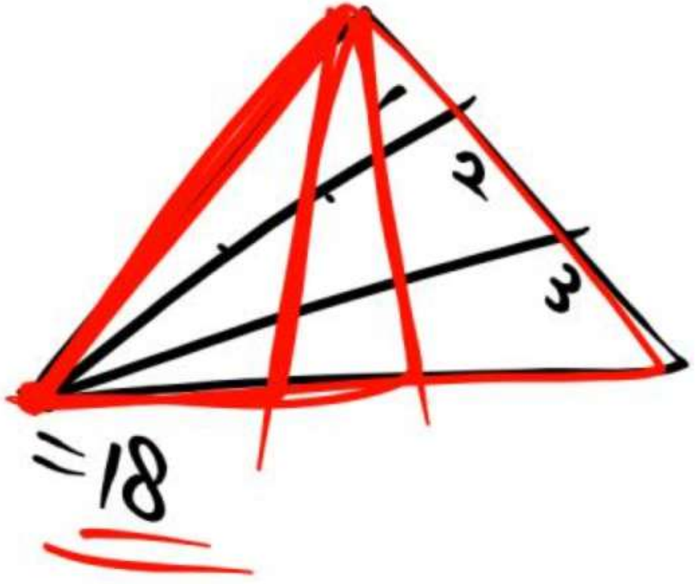
$3^3 = 27$ Ans

18+18−9

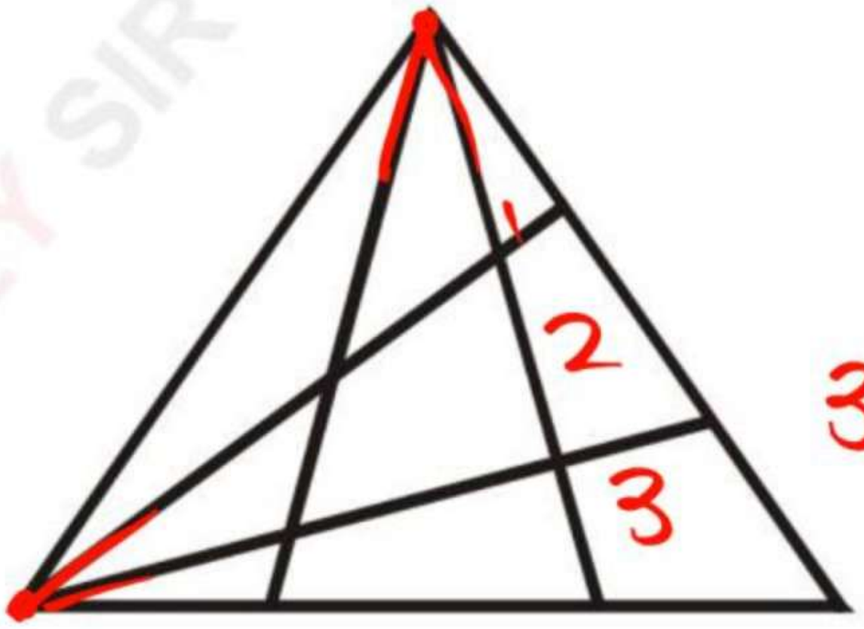
= 36−9

= 27





10.



$3^3 = 27$

Find out the number of triangles in the given figure?

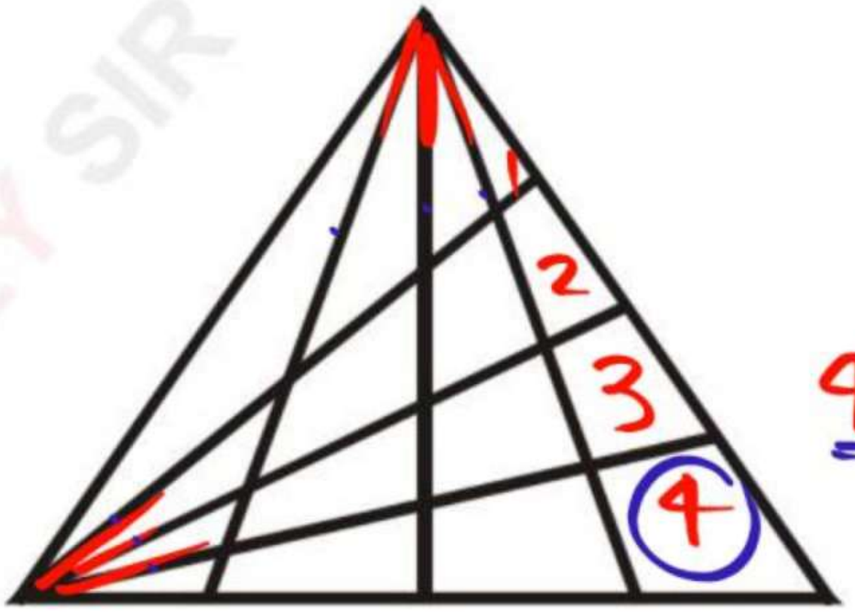
दी गई आकृति में त्रिभुजों की संख्या बताइए?

(a) 25 **(b) 30**

(c) 18 **(d) 27**

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11.



$4^3 = 64$

Find out the number of triangles in the given figure?

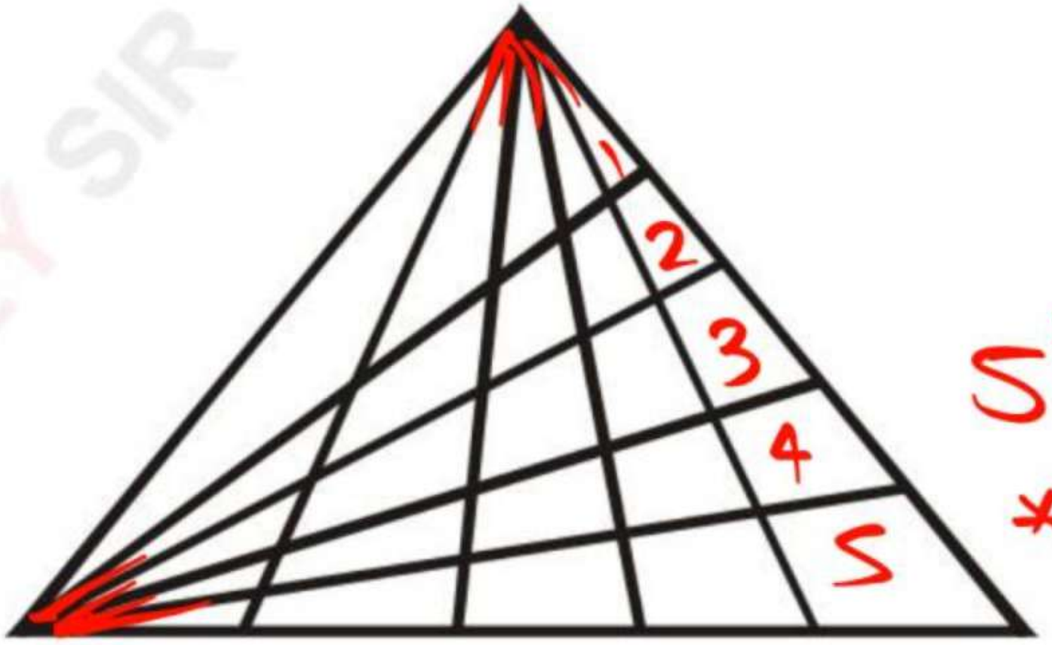
दी गई आकृति में त्रिभुजों की संख्या बताइए?

(a) 64 **(b) 60**

(c) 72 **(d) 70**

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12.



$S^3 = 125$

*TC

Find out the number of triangles in the given figure?

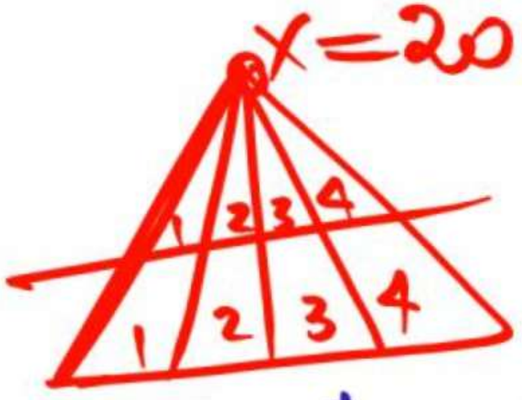
दी गई आकृति में त्रिभुजों की संख्या बताइए?

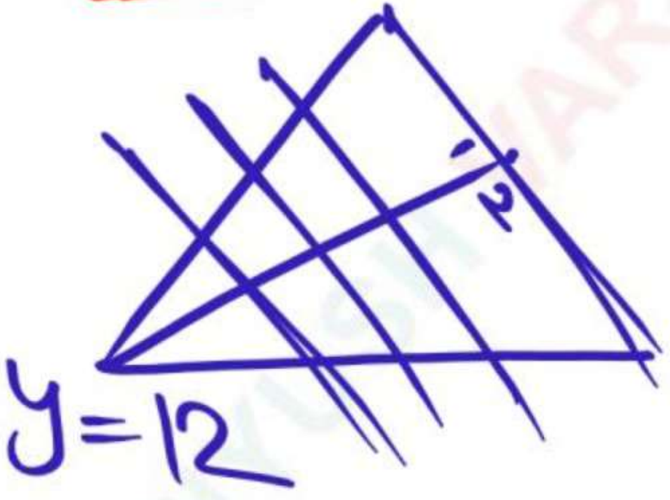
(a) 120 (b) 125

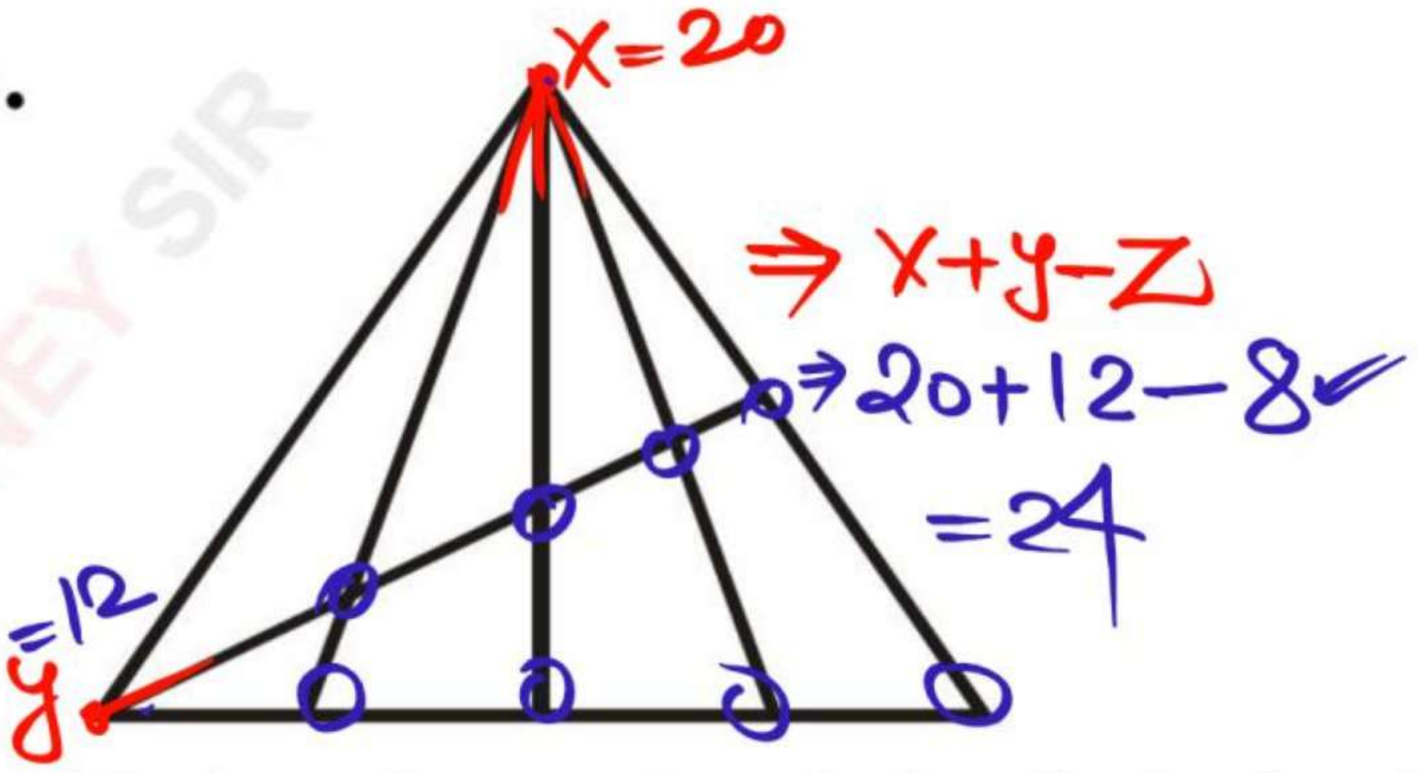
(c) 115 (d) 130

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13.







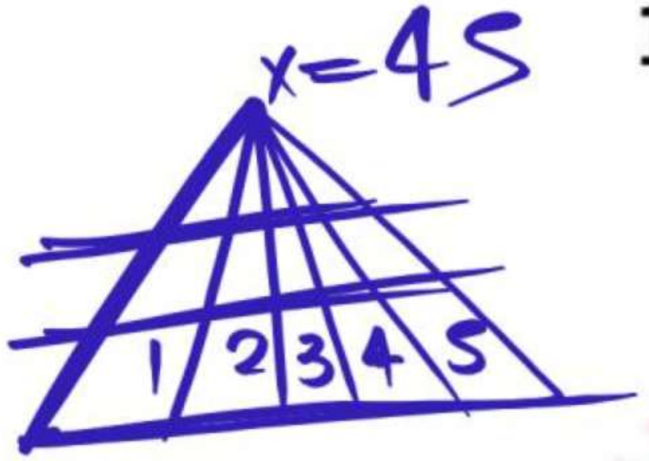
Find out the number of triangles in the given figure?

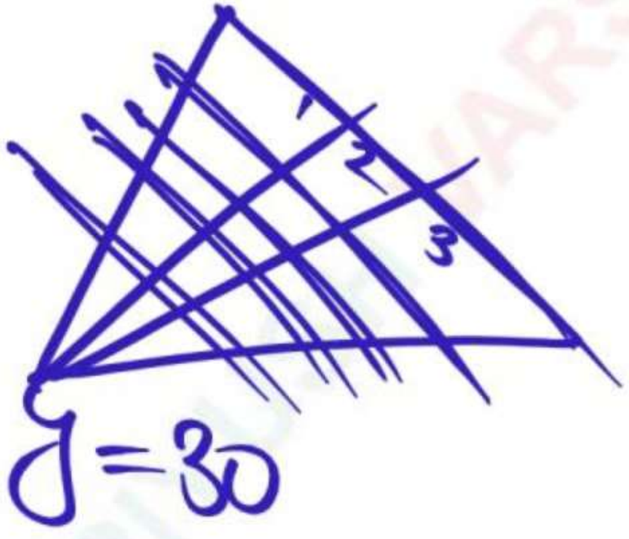
दी गई आकृति में त्रिभुजों की संख्या बताइए?

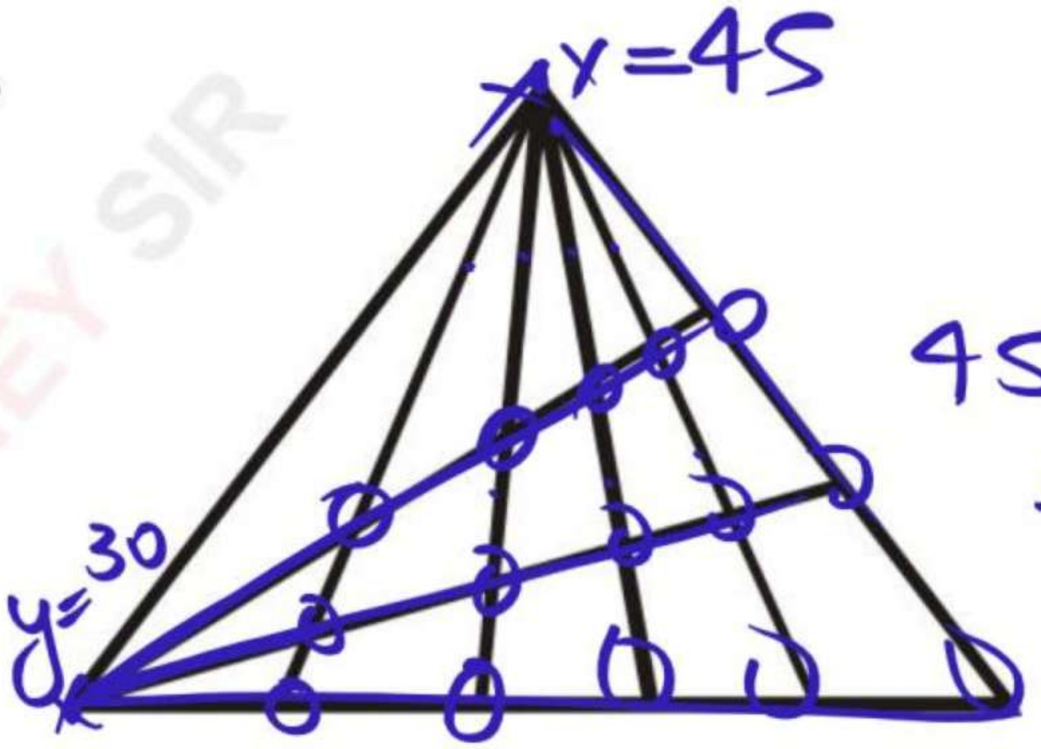
(a) 24
(b) 28
(c) 30
(d) 20

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14.







45+30−15
=60

Find out the number of triangles in the given figure?

दी गई आकृति में त्रिभुजों की संख्या बताइए?

(a) 50 **(b) 60**

(c) 70 **(d) 80**

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$\Delta = 1+2+3+4$

$\Delta = 10$

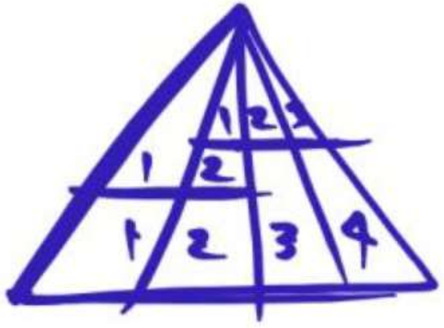
①



$\Delta = 10 \times 3$

$= 30$

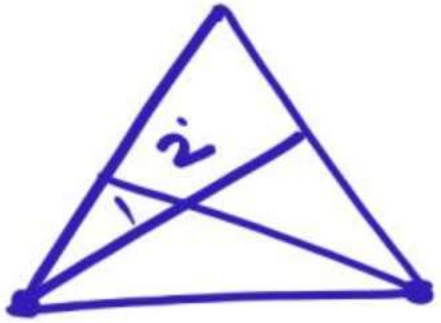
②



$\Delta = 10 + 3 + 6$

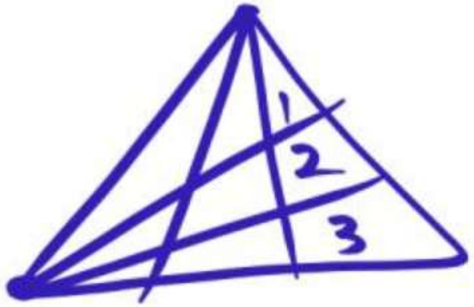
$\Delta = 19$

③



$\Delta = 2^3$

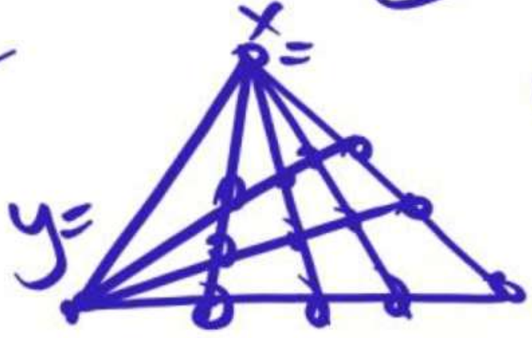
$= 8$



$\Delta = 3^3$

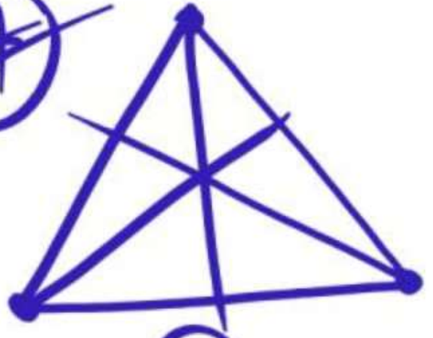
$= 27$

④



$\Delta = X+Y-Z$

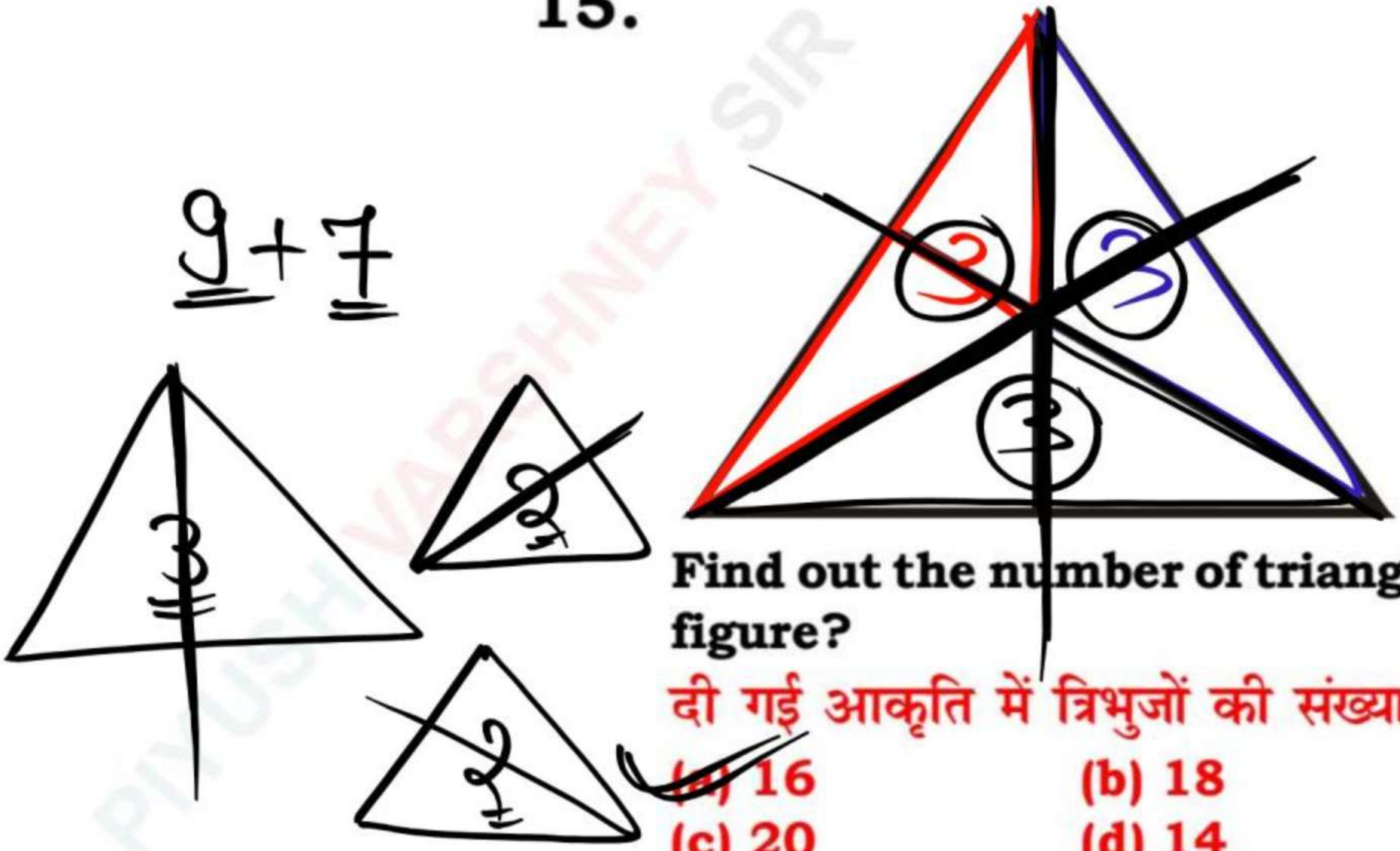
⑤



⑥



15.



Find out the number of triangles in the given figure?

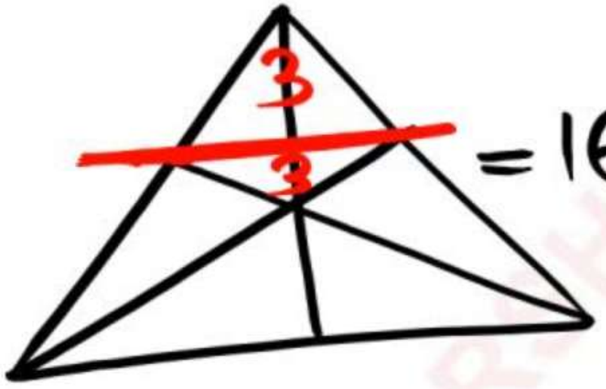
दी गई आकृति में त्रिभुजों की संख्या बताइए?

(a) 16
(b) 18
(c) 20
(d) 14

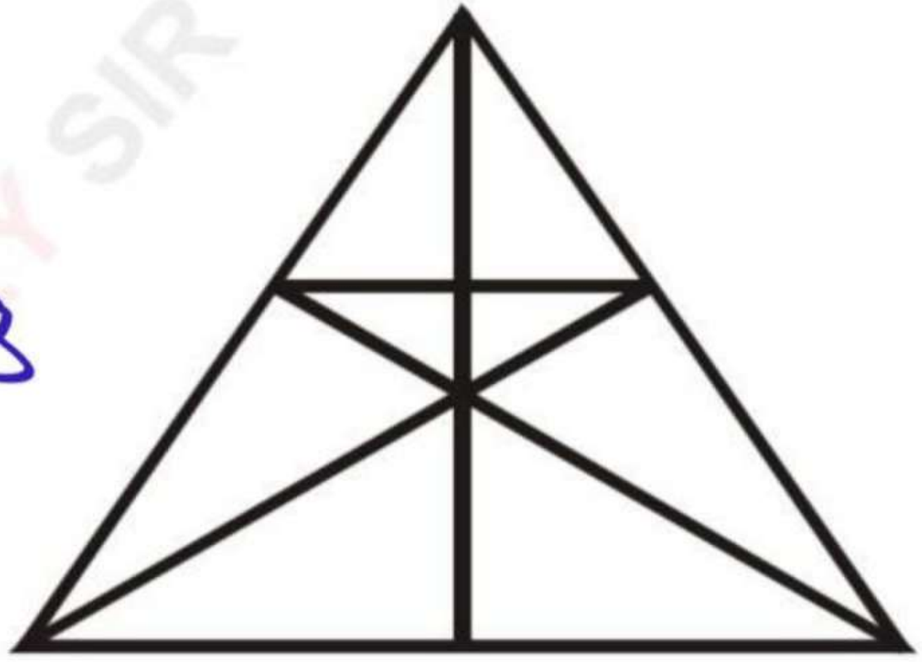
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Piyush Varshney Sir (Reasoning Expert)

16.



= 16 + 8



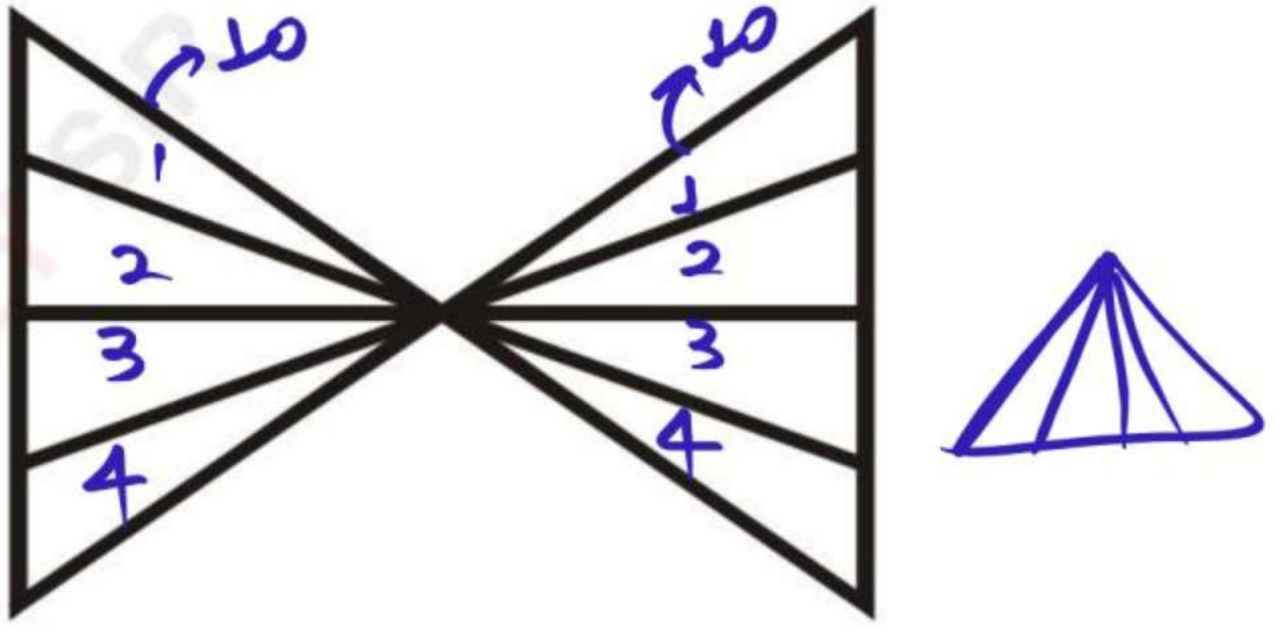
Find out the number of triangles in the given figure?

दी गई आकृति में त्रिभुजों की संख्या बताइए?

(a) 15
(b) 14
(c) 13
(d) 24

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17.



Single figure

Multiple figures

अपनी आखों + मन की आँखों

Basic figure + Extra

Find out the number of triangles in the given figure?

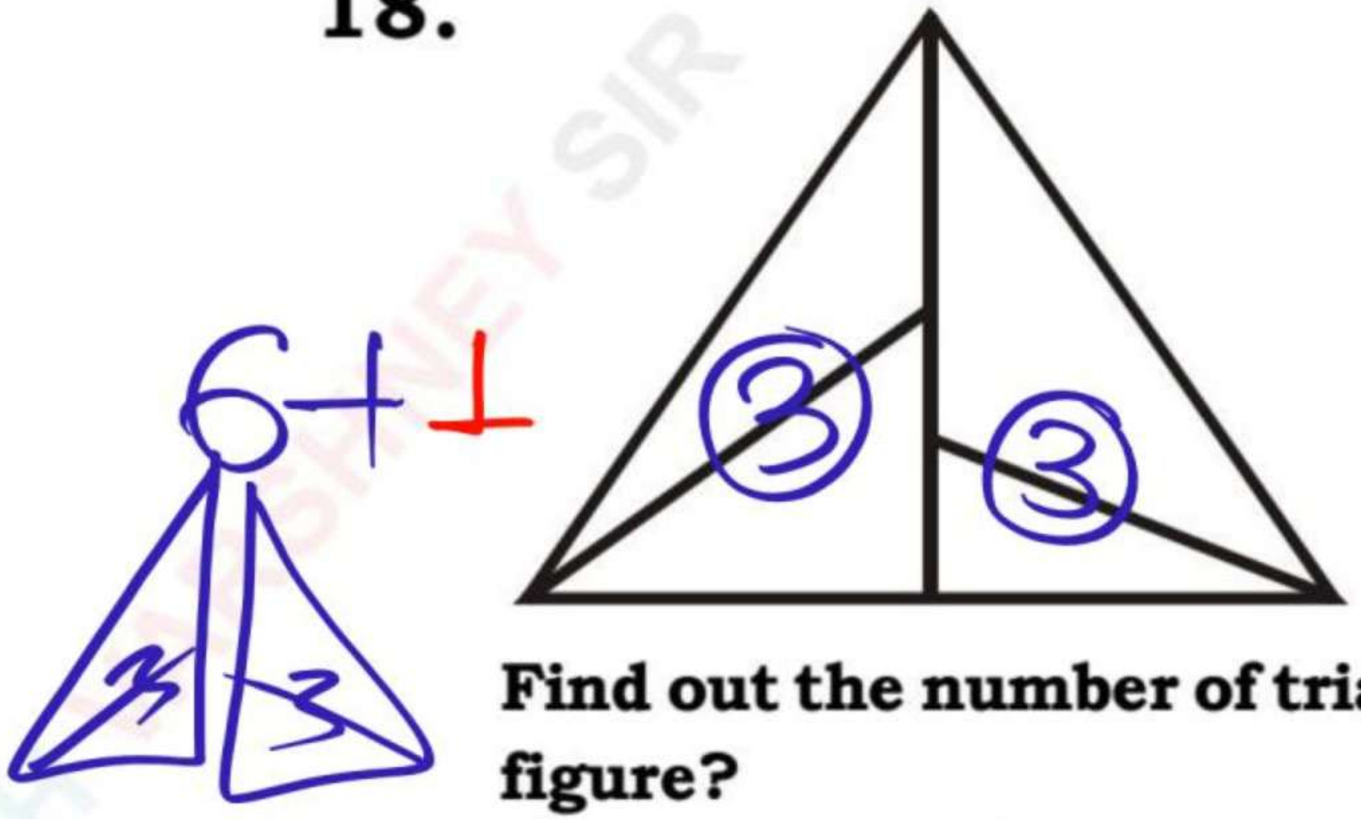
दी गई आकृति में त्रिभुजों की संख्या बताइए?

(a) 18 **(b) 22**

(c) 20 10+10 **(d) 24**

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18.



Find out the number of triangles in the given figure?

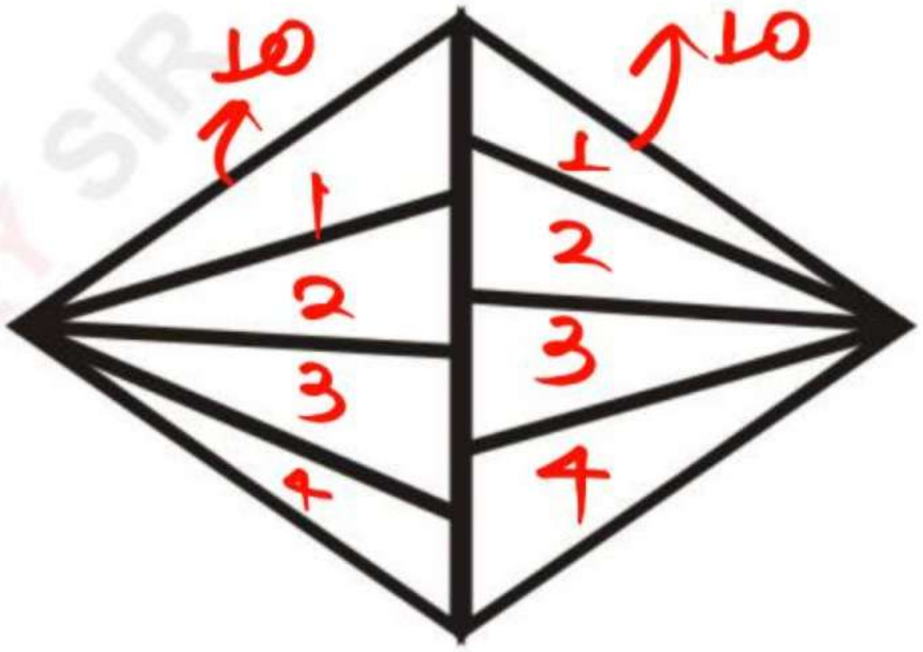
दी गई आकृति में त्रिभुजों की संख्या बताइए?

(a) 6 **(b) 8**

(c) 7 **(d) 9**

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19.



Find out the number of triangles in the given figure?

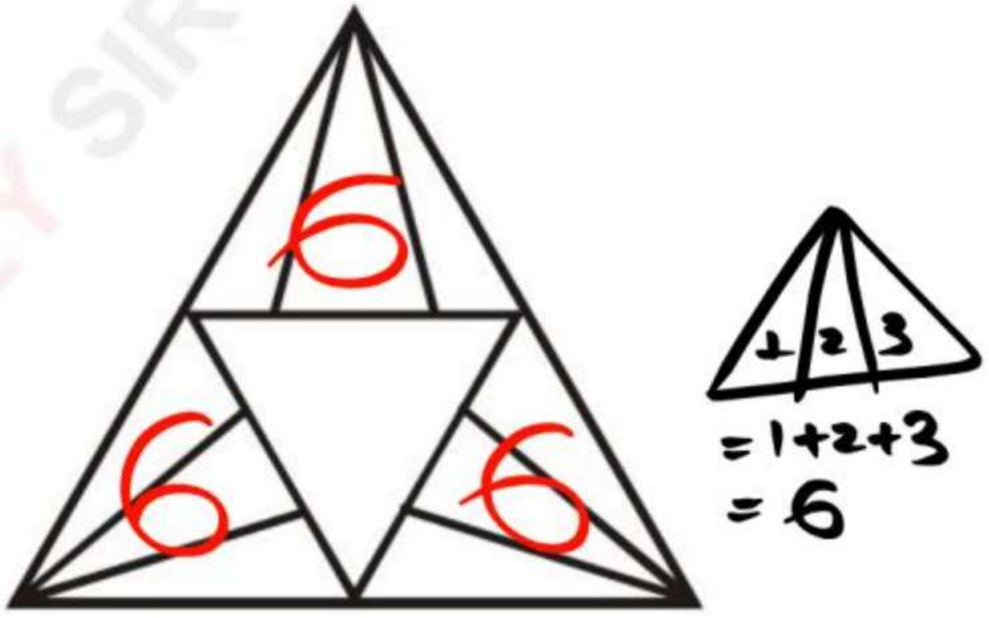
दी गई आकृति में त्रिभुजों की संख्या बताइए?

(a) 18 **(b) 19**

(c) 20 **(d) 21**

PIYUSH VARSHNEY SIR

20.



18+2
=20

=1+2+3
=6

Find out the number of triangles in the given figure?

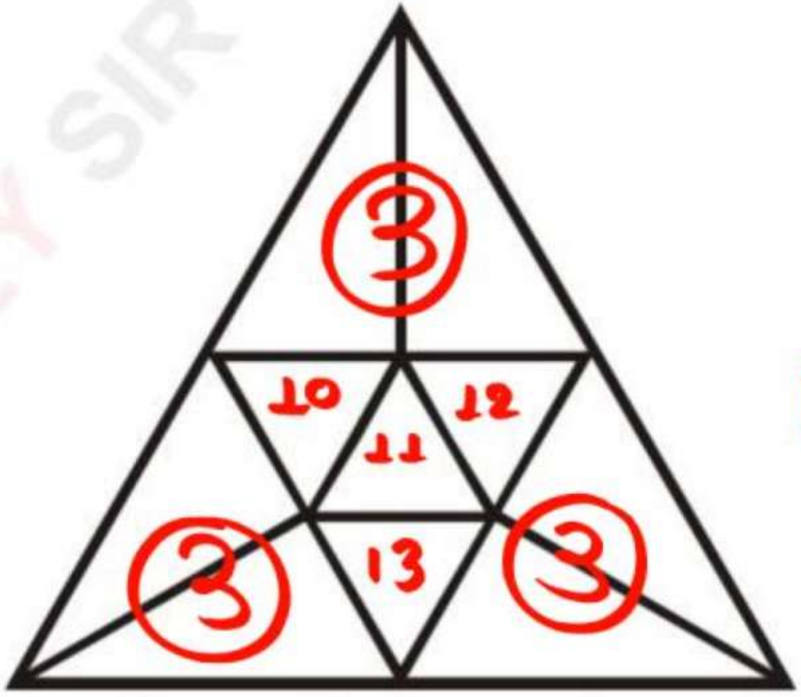
दी गई आकृति में त्रिभुजों की संख्या बताइए?

(a) 20 **(b) 18**

(c) 22 **(d) 19**

PIYUSH VARSHNEY SIR

21.



1st time
= Time

2nd time
= 10sec

13 + 2

Find out the number of triangles in the given figure?

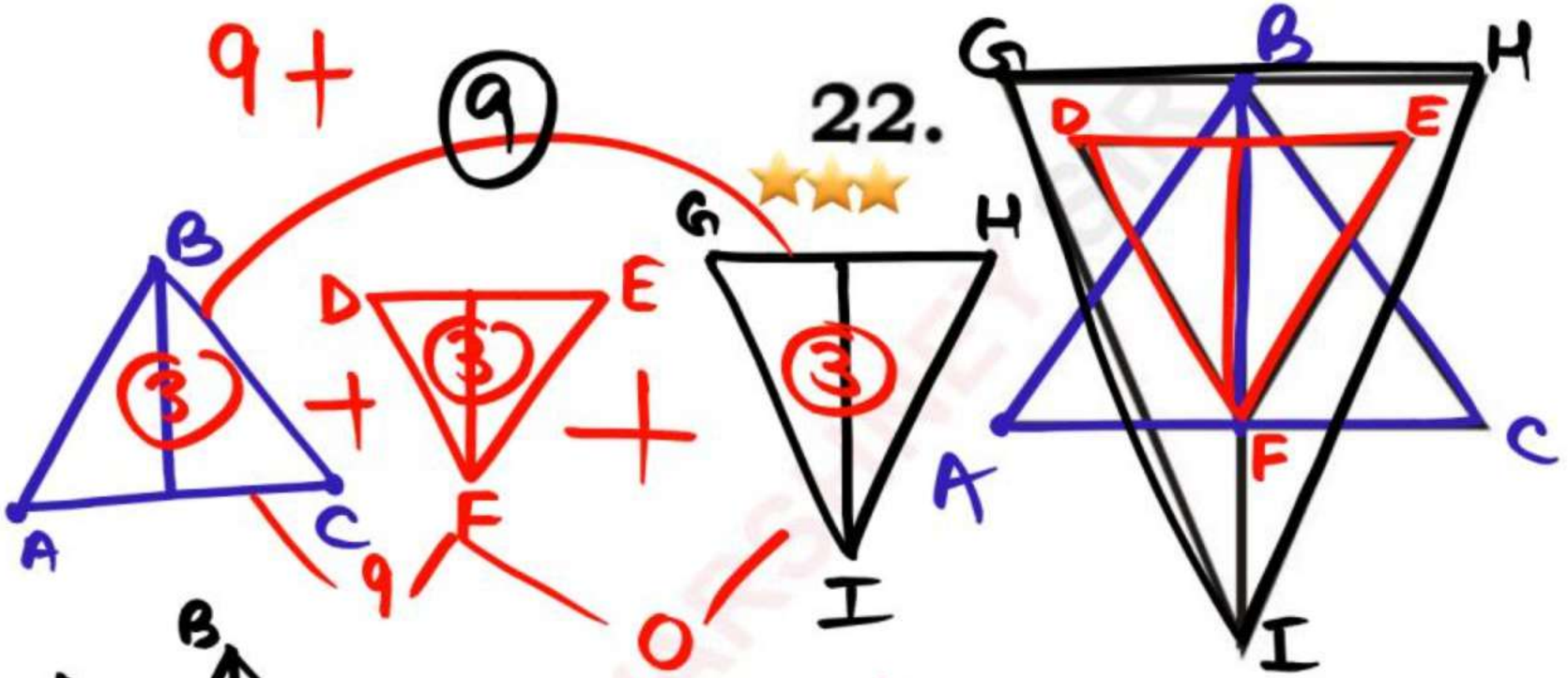
दी गई आकृति में त्रिभुजों की संख्या बताइए?

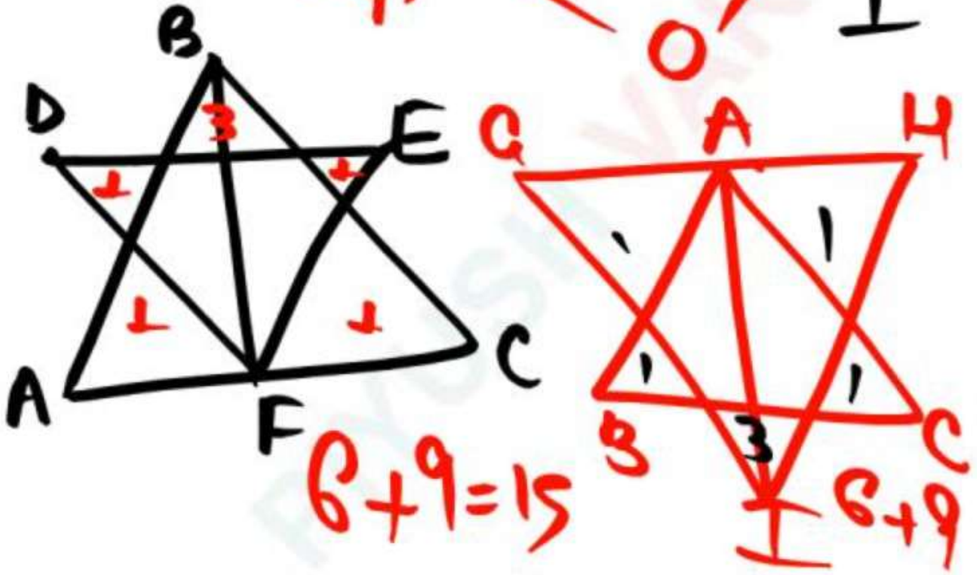
(a) 20 **(b) 15**

(c) 16 **(d) 18**

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22.





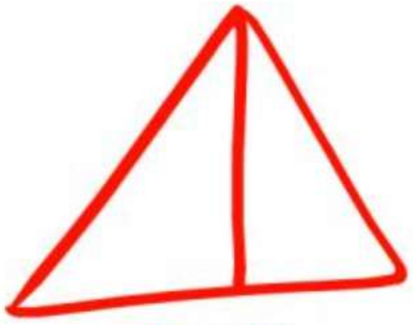
Find out the number of triangles in the given figure?

दी गई आकृति में त्रिभुजों की संख्या बताइए?

(a) 27 (b) 25

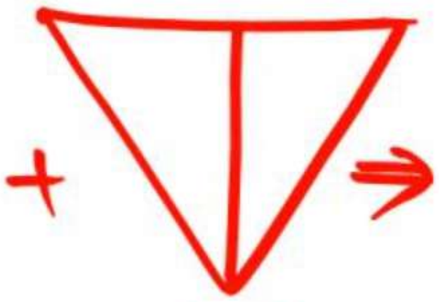
(c) 30 (d) 32

Piyush Varshney Sir (Reasoning Exper



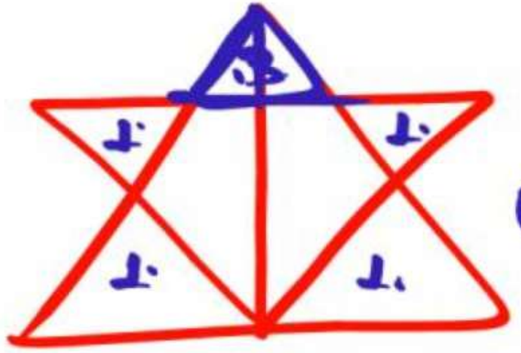
$\Delta = 3$

+



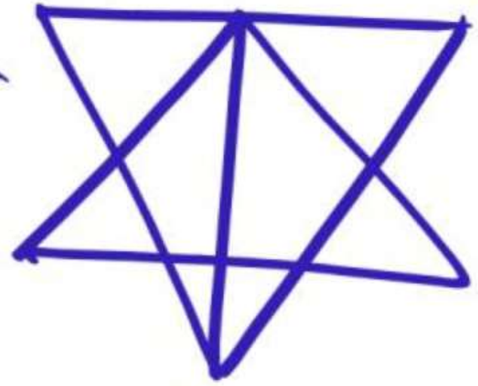
$\Delta = 3$

⇒

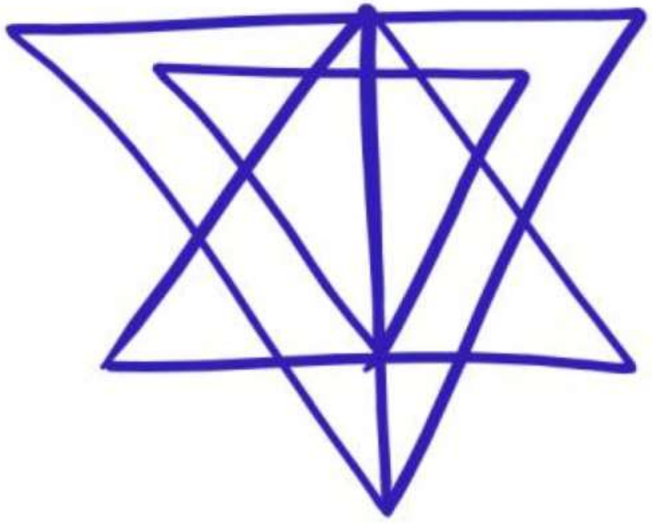


$6 + 9 = 15$

or



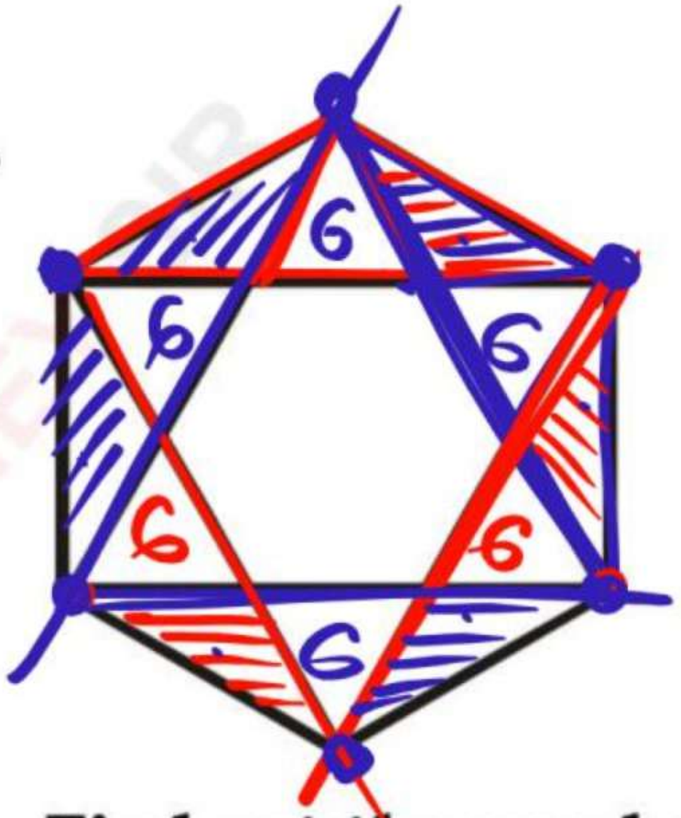
$6 + 9 = 15$



$9 + 18$

23.

★★



36−6+2



(कल)
=6

Find out the number of triangles in the given figure?

दी गई आकृति में त्रिभुजों की संख्या बताइए?

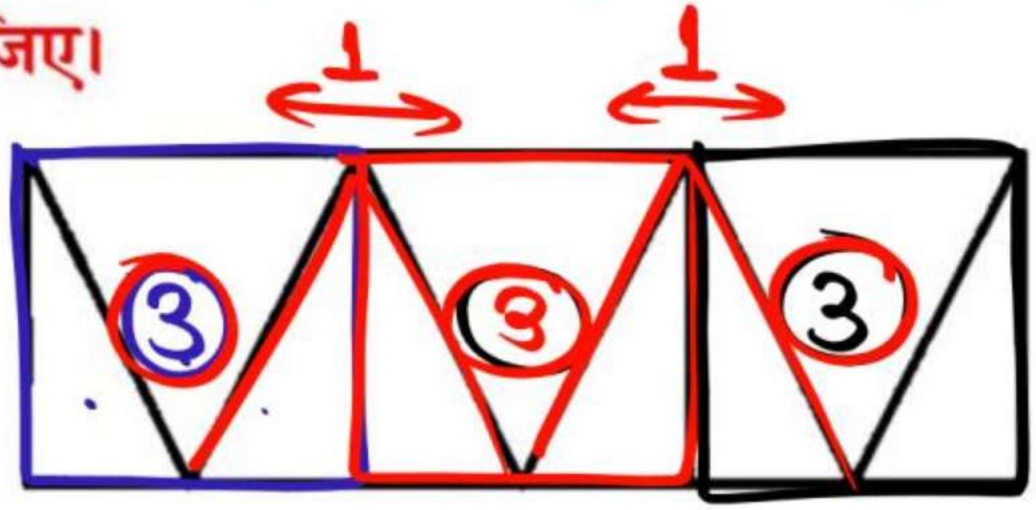
(a) 32 (b) 36

(c) 34 (d) 28

PIYUSH VARSHNEY GROUP

24. Find the total number of triangles in the given figure.

दी गई आकृति में त्रिभुजों की कुल संख्या ज्ञात कीजिए।



~~(a) 11~~

(b) 9

(c) 10

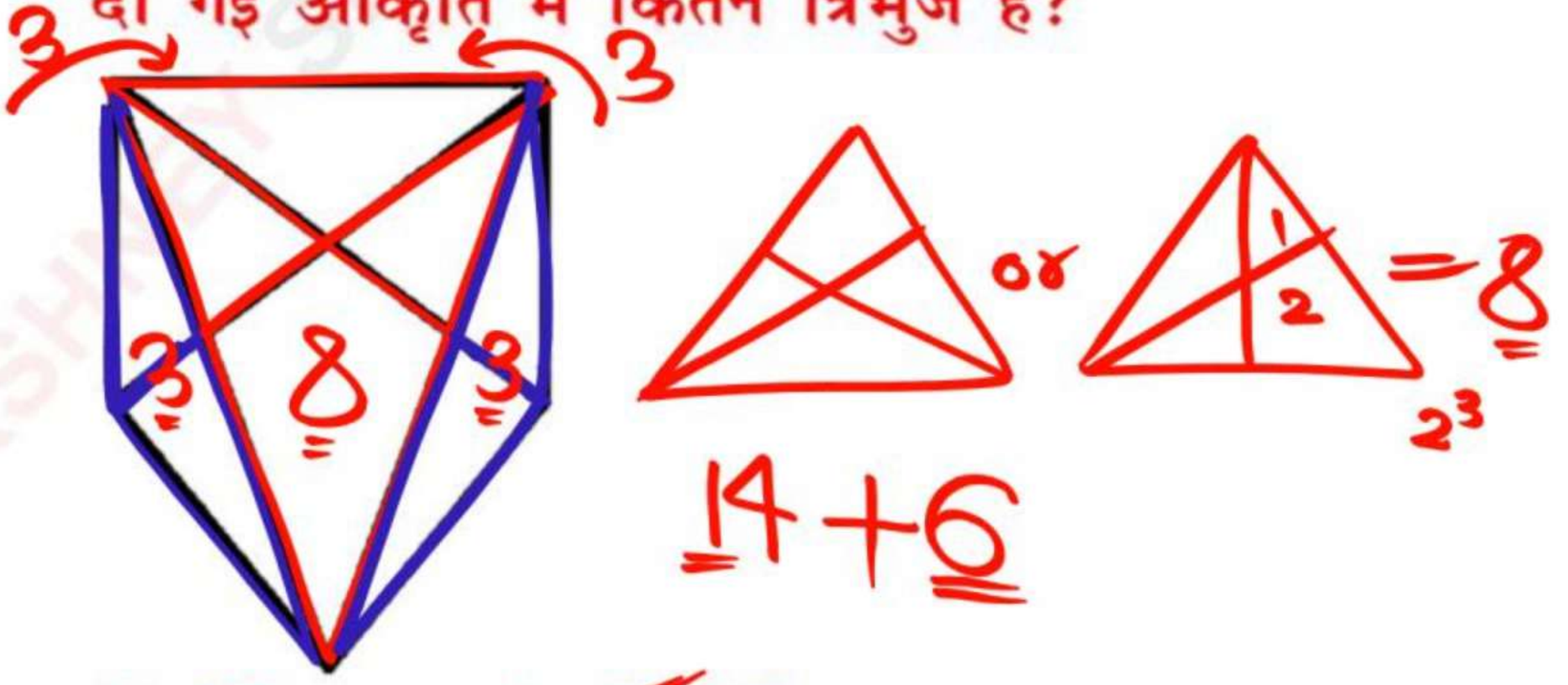
(d) 10

9+2=11

PIYUSH VARSHNEY SIR

★★★ **25.** **How many triangles are there in the given figure?**

दी गई आकृति में कितने त्रिभुज हैं?



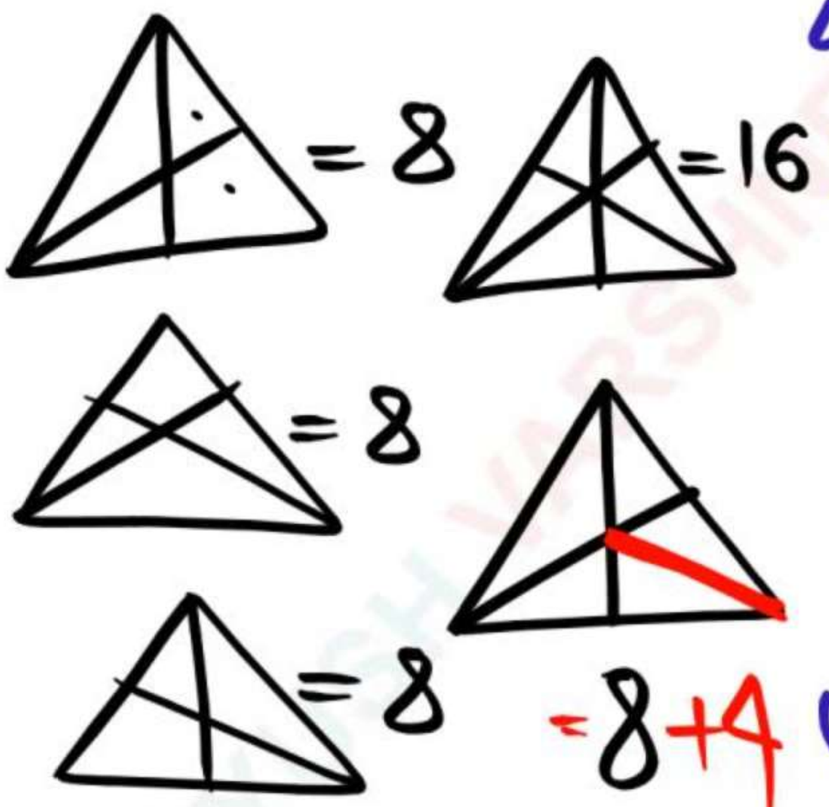
(a) 19 (b) 20

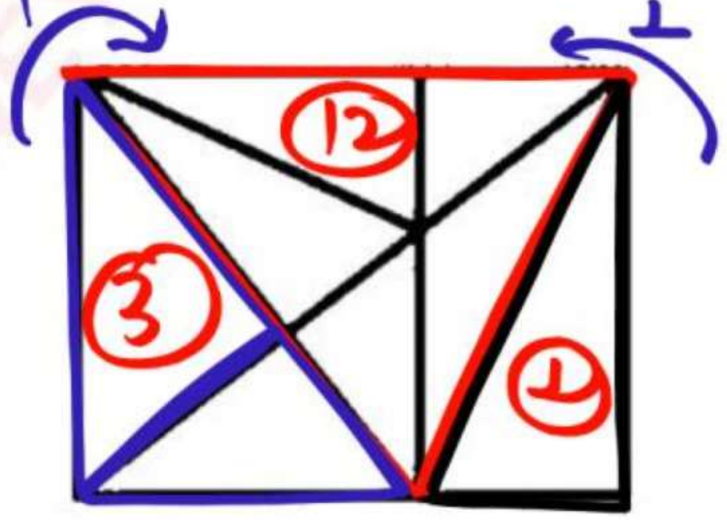
(c) 22 (d) 21

PIYUSH VARSHNEY SIR

26. **How many triangles are there in the figure given below?**

नीचे दी गई आकृति में कितने त्रिभुज हैं?





16 + 5

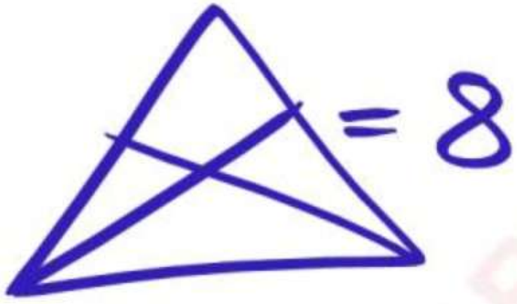
(a) 22 (b) 19

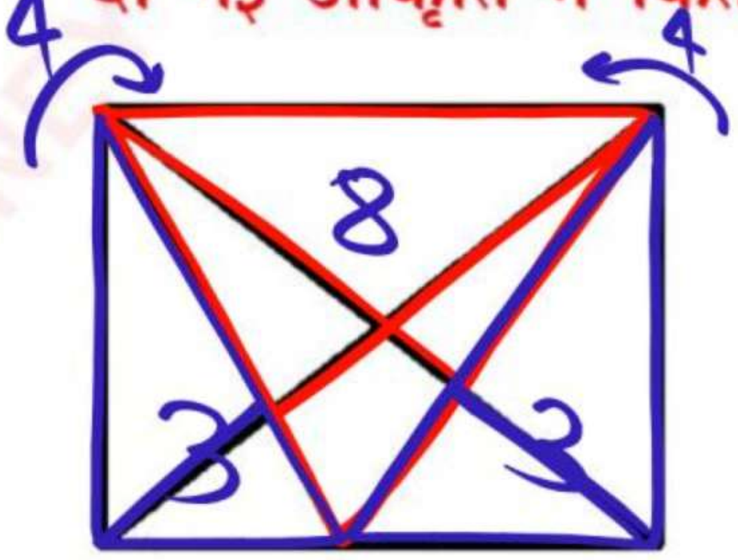
(c) 21 (d) 20

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27. **How many triangles are there in the given figure?**

दी गई आकृति में कितने त्रिभुज हैं?





14 + 8 + 1

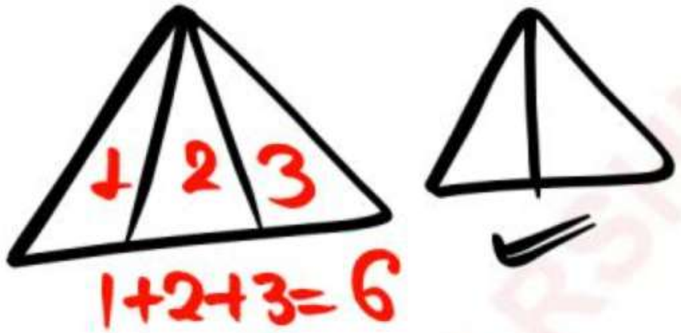
(a) 22 (b) 20

(c) ~~23~~ (d) 21

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28.

Type-2

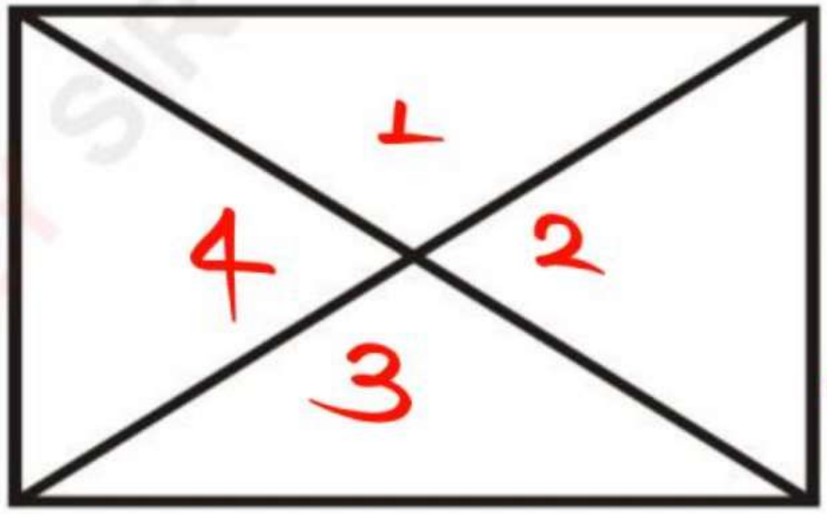


1+2+3=6



=4×2

=8



Basic

Δ=1, 2, 3, 4, 12, 23, 34, 41 = 8

4×2=8

Find out the number of triangles in the given figure?

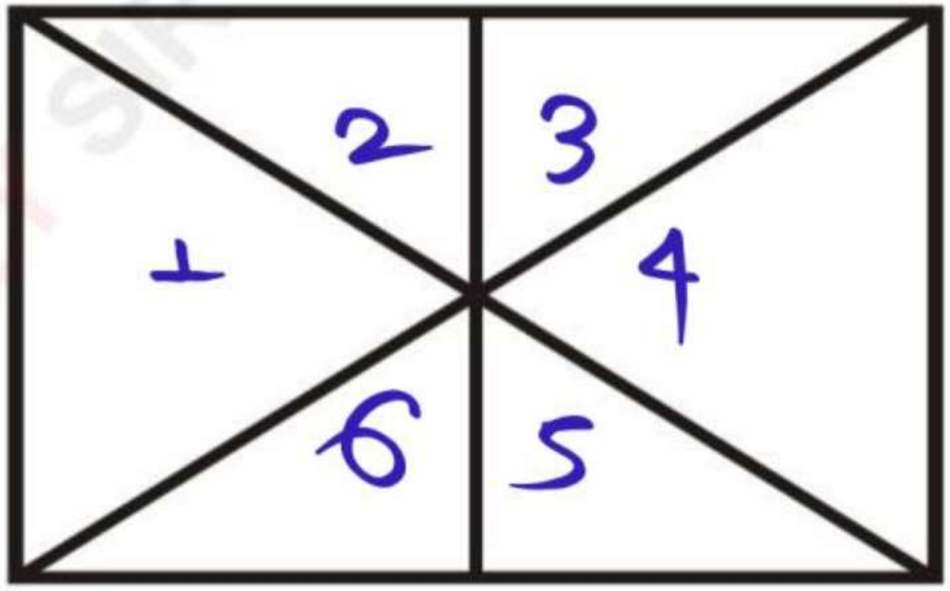
दी गई आकृति में त्रिभुजों की संख्या बताइए?

(a) 8 **(b) 10**

(c) 7 **(d) 6**

PIYUSH VARSHNEY SIR

29.



$\triangle = 6 \times 2$
$= 12$

Find out the number of triangles in the given figure?

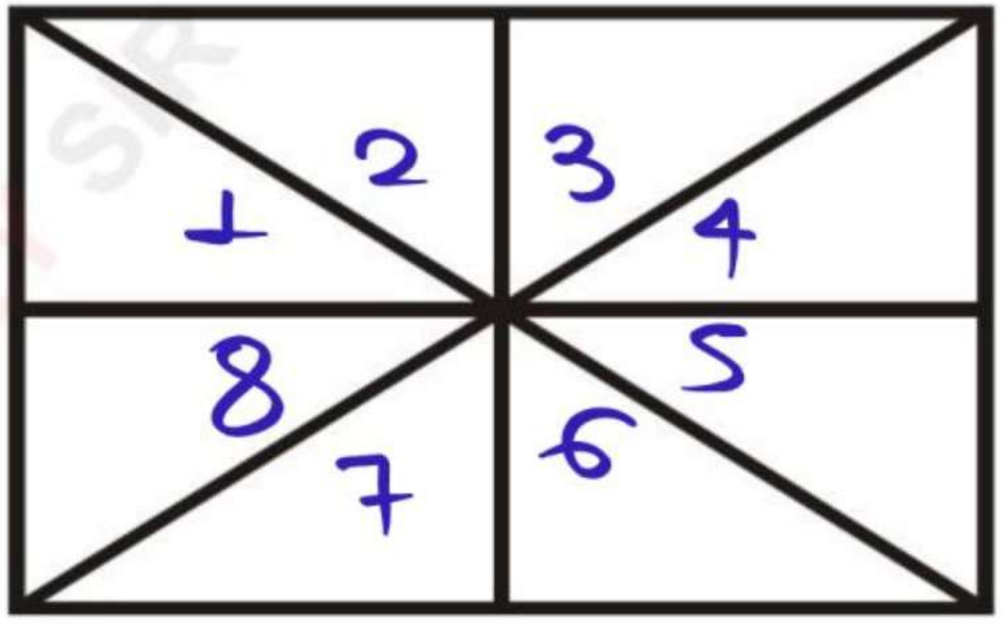
दी गई आकृति में त्रिभुजों की संख्या बताइए?

(a) 11 **(b) 10**

(c) 12 **(d) 14**

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30.



Find out the number of triangles in the given figure?

दी गई आकृति में त्रिभुजों की संख्या बताइए?

(a) 20 **(b) 14**

(c) 18 **(d) 16**

$\Delta = 8 \times 2$

$= 16$

Basic

1, 2, 3, 4, 5, 6, 7, 8

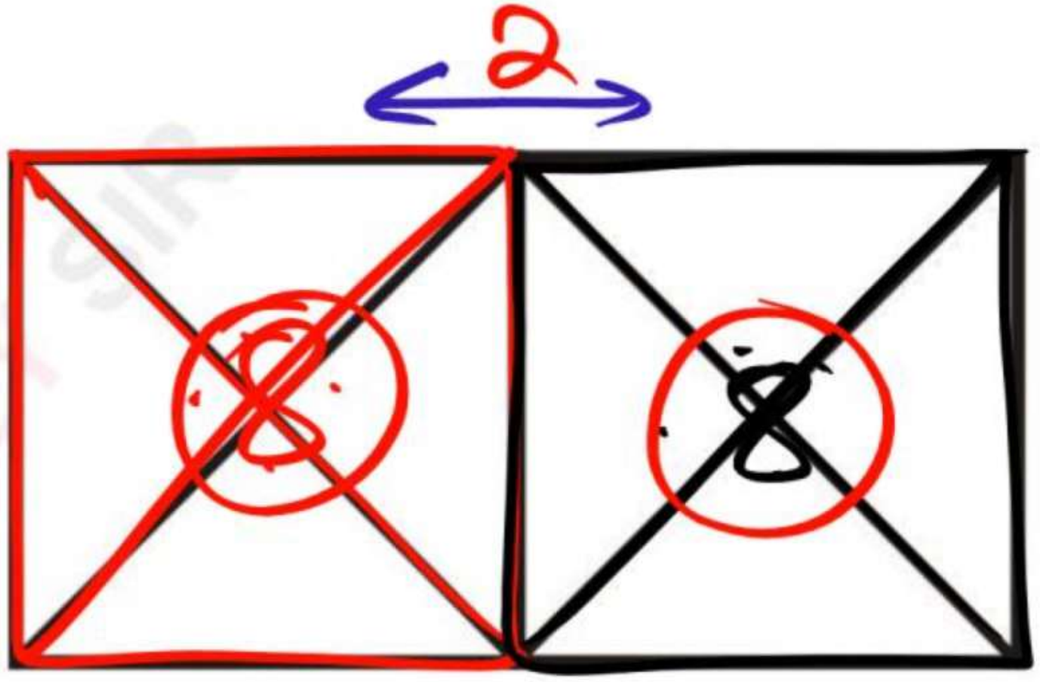
23, 45, 67, 81

8123, 2345, 4567, 6781

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Basic + Extra
16 + 2

31.



Find out the number of triangles in the given figure?

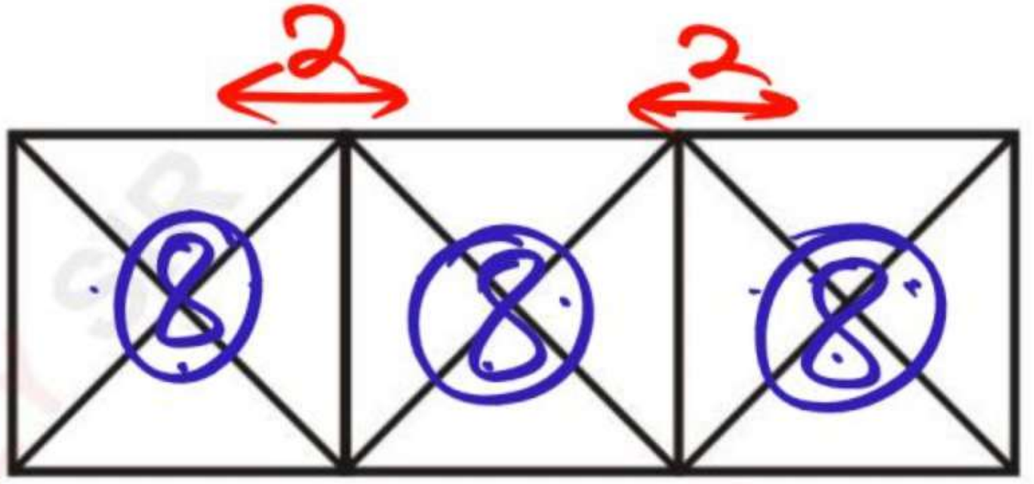
दी गई आकृति में त्रिभुजों की संख्या बताइए?

(a) 16 **(b) 18**

(c) 17 **(d) 20**

PIYUSH VARSHNEY SIR

32



24 + 4

Find out the number of triangles in the given figure?

दी गई आकृति में त्रिभुजों की संख्या बताइए?

(a) 20

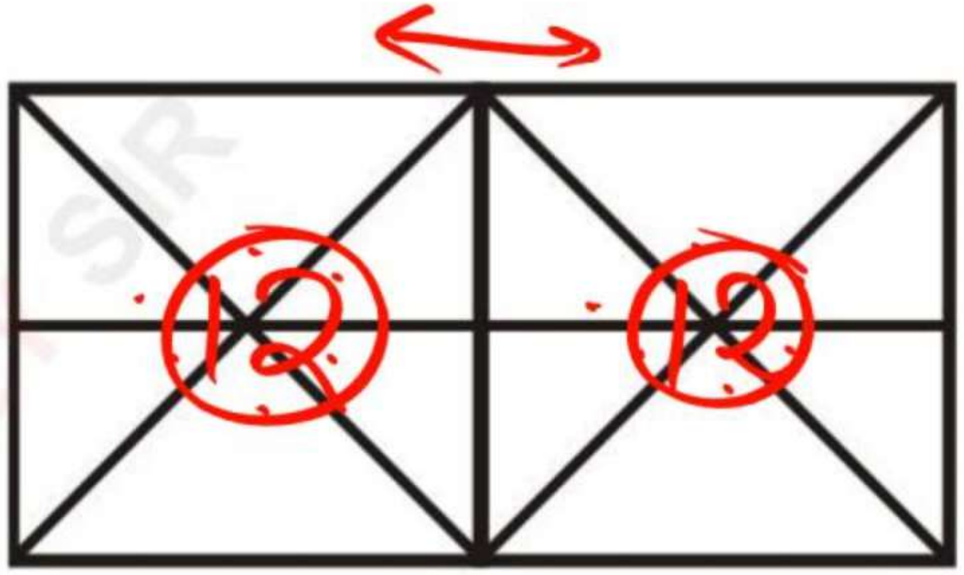
(b) 24

(c) 28

(d) 32

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33.



24+4

=28

Find out the number of triangles in the given figure?

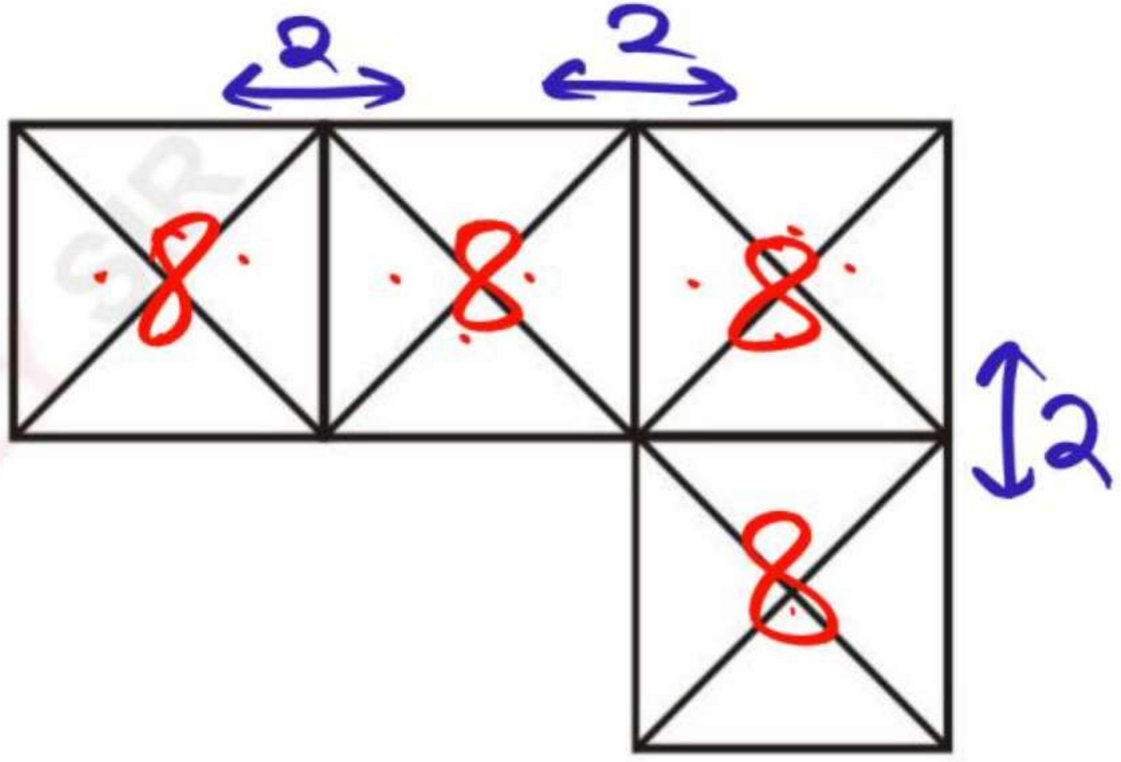
दी गई आकृति में त्रिभुजों की संख्या बताइए?

~~(a) 28~~ (b) 24

(c) 27 (d) 25

PIYUSH VARSHNEY SIR

34.



32+6+1

Find out the number of triangles in the given figure?

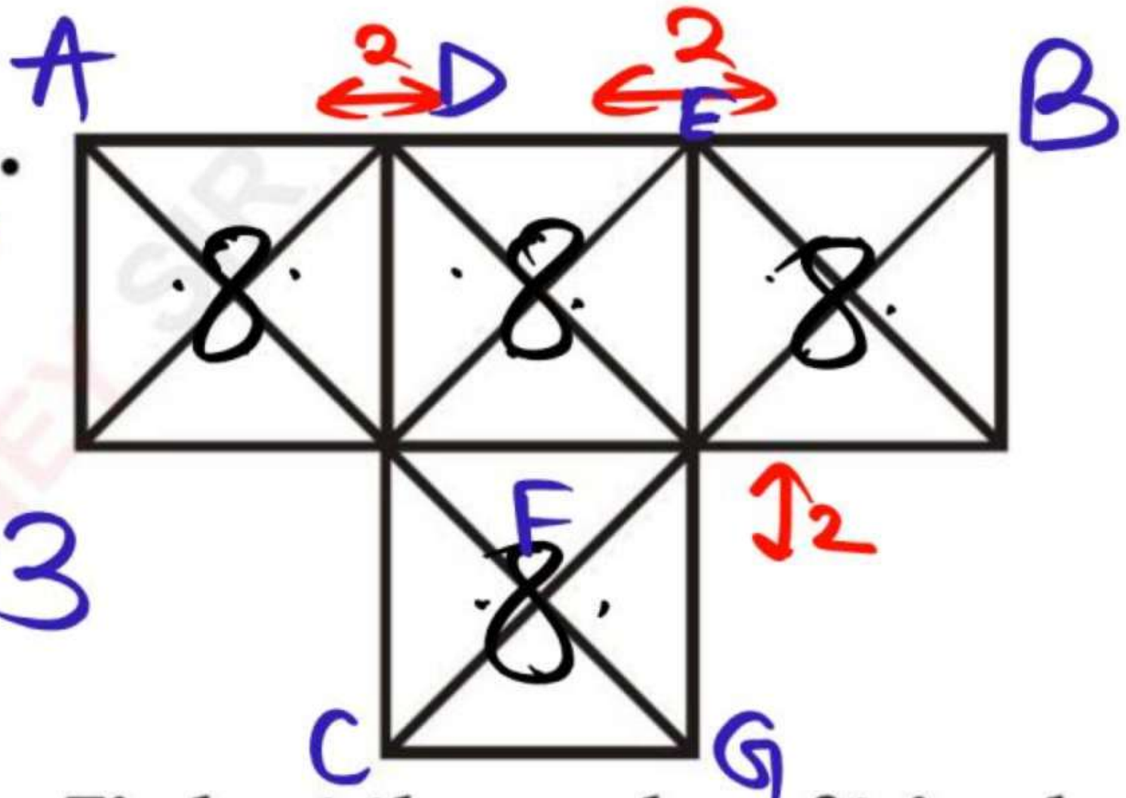
दी गई आकृति में त्रिभुजों की संख्या बताइए?

(a) 35 **(b) 32**

(c) 38 **(d) 39**

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35. ★★★



32+6+3

ABF
AEG
BDC

Find out the number of triangles in the given figure?

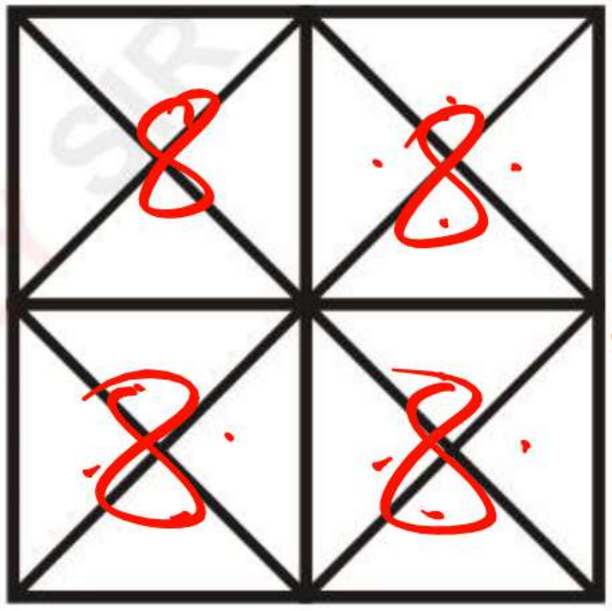
दी गई आकृति में त्रिभुजों की संख्या बताइए?

(a) 40 **(b) 41**

(c) 42 **(d) 45**

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36.



Find out the number of triangles in the given figure?

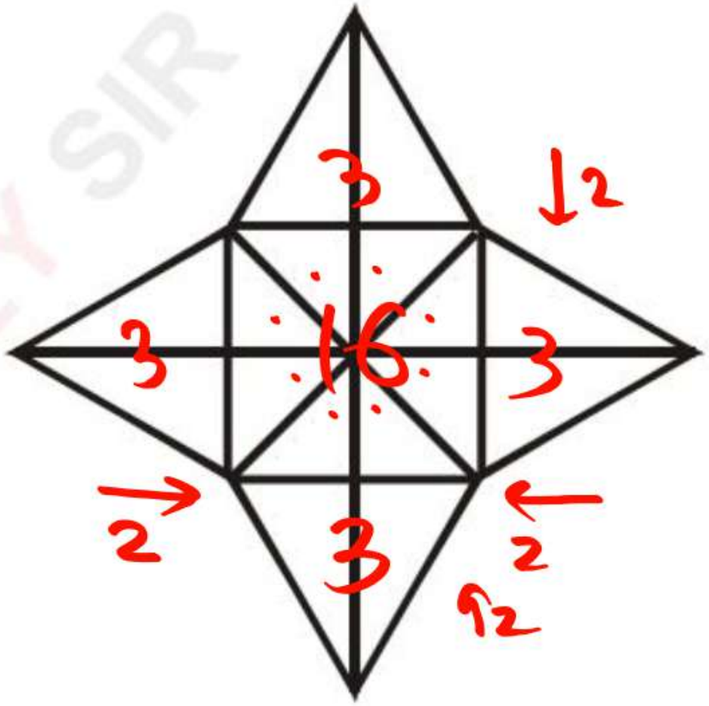
दी गई आकृति में त्रिभुजों की संख्या बताइए?

(a) 36 (b) 38

(c) 40 or more (d) 34

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37.



16+12+8=36

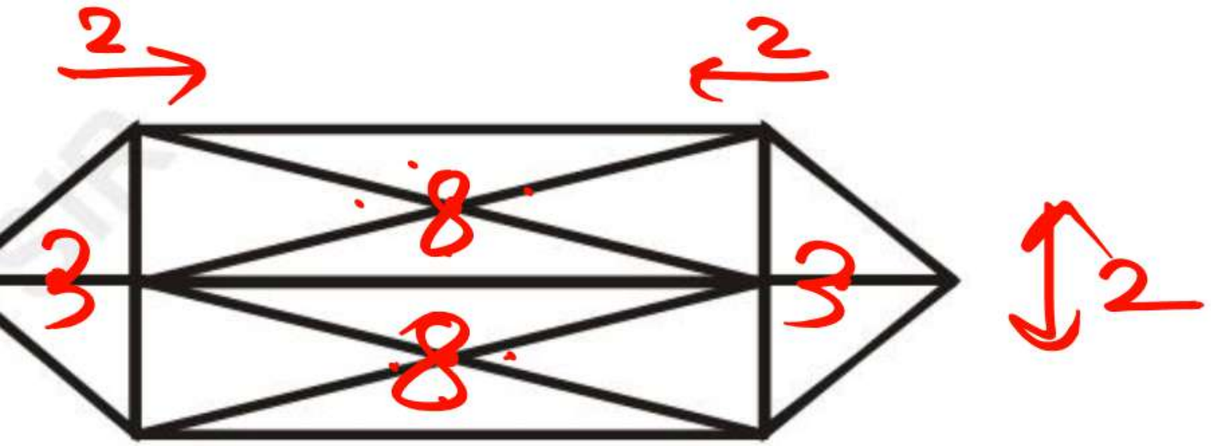
Find out the number of triangles in the given figure?

दी गई आकृति में त्रिभुजों की संख्या बताइए?

(a) 36 (b) 40

(c) 34 (d) 30

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nd out the number of triangles in the ven figure?

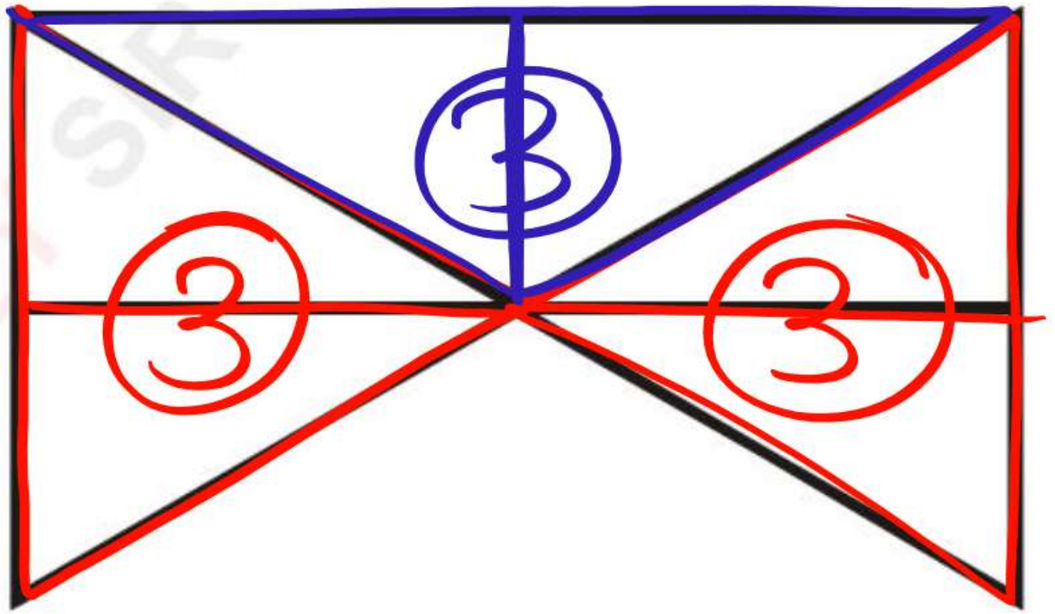
गई आकृति में त्रिभुजों की संख्या बताइए?

) 22
) 24
) 28
) 32

22+6

9+2
=11

39.



Find out the number of triangles in the given figure?

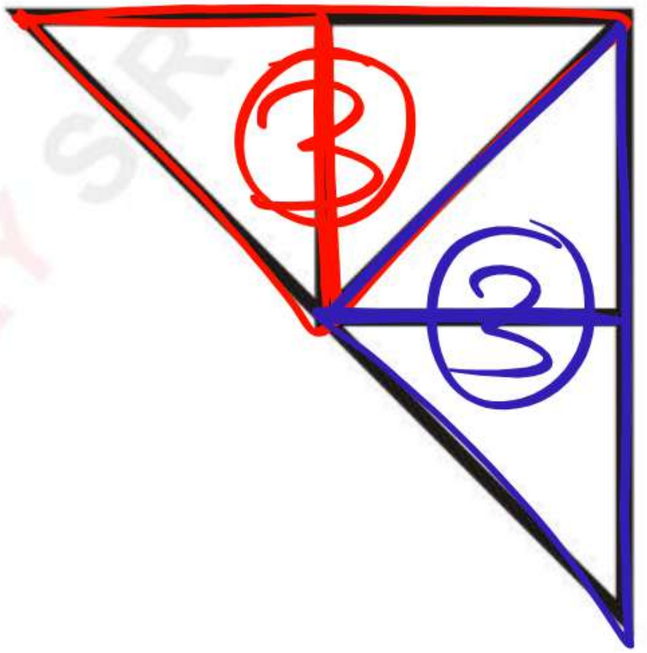
दी गई आकृति में त्रिभुजों की संख्या बताइए?

(a) 10 **(b) 11**

(c) 9 **(d) 12**

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40.



6 + 1

Find out the number of triangles in the given figure?

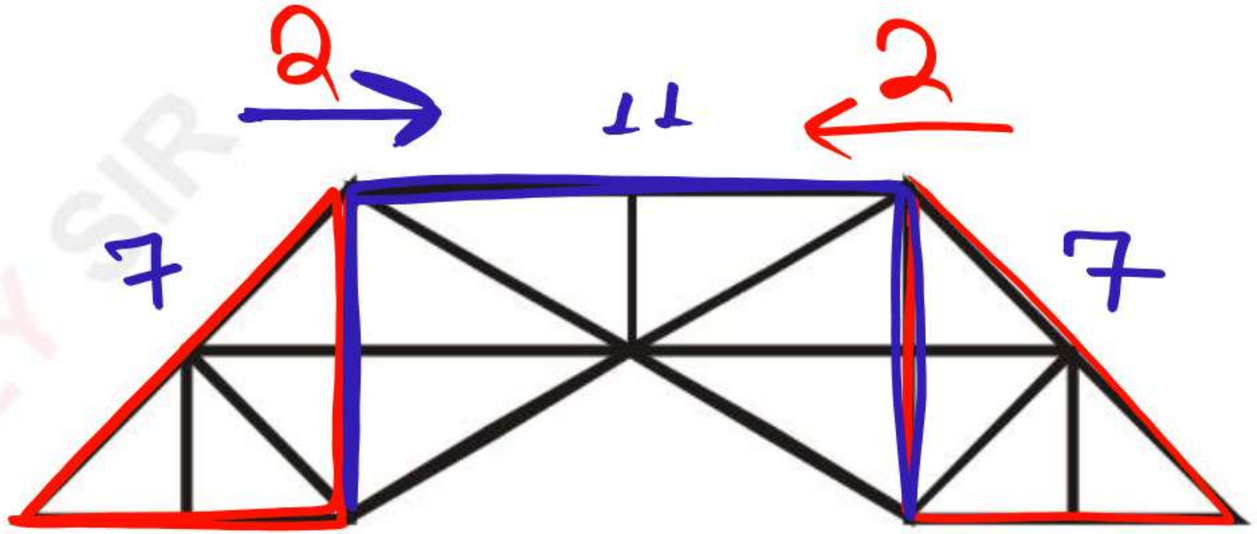
दी गई आकृति में त्रिभुजों की संख्या बताइए?

(a) 5 **(b) 10**

(c) 6 **(d) 7**

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41.



25+4

Find out the number of triangles in the given figure?

दी गई आकृति में त्रिभुजों की संख्या बताइए?

(a) 33

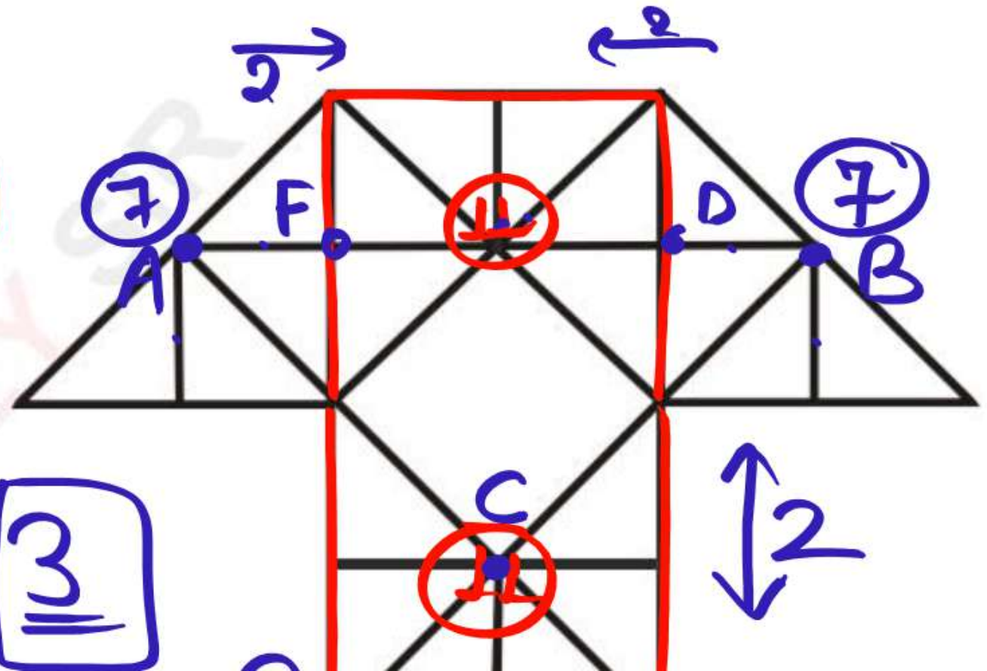
(b) 29

(c) 28

(d) 31

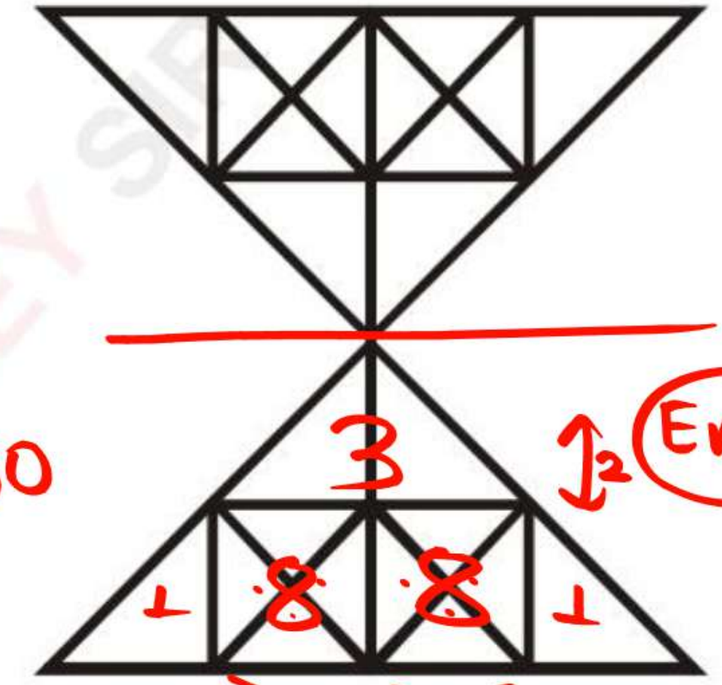
PIYUSH VARSHNEY SIR

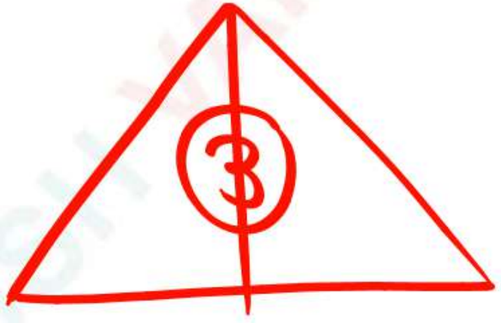
42.



$36 + 6 + \boxed{3}$

43.





Find out the number of triangles in the given figure?

दी गई आकृति में त्रिभुजों की संख्या बताइए?

(a) 60 (b) 70

(c) 45 (d) 75

(Counting Exam में (options)

Basic Extra

$\Delta = 21 + 9 = 30$

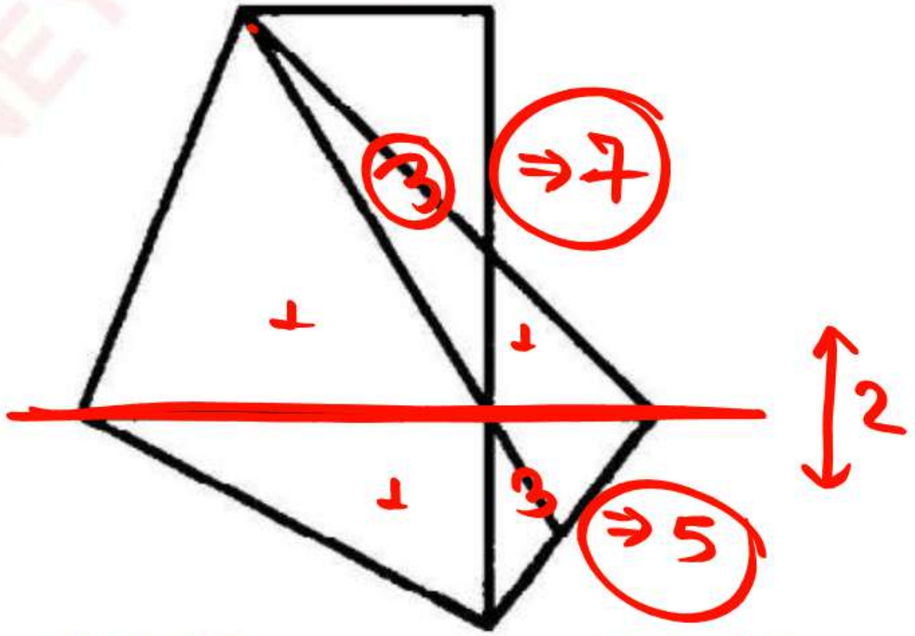
45.

75.

PIYUSH VARSHNEY SIR

44. **How many triangles are there in the given figure?**

दी गई आकृति में कितने त्रिभुज हैं?



(a) 13 (b) 12

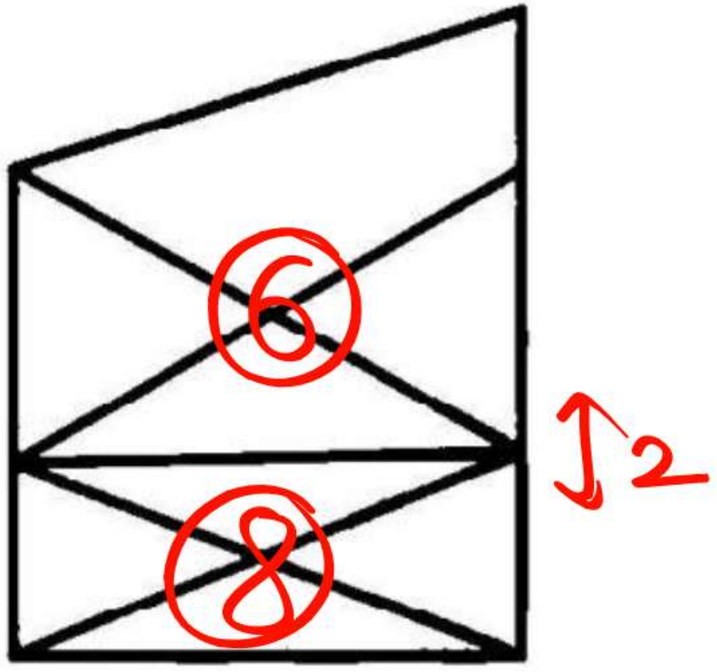
(c) 11 (d) 14

PIYUSH VARSHNEY SIR

Piyush Varshney Sir (Reasoning Expert)

45. **How many triangles are there in the given figure?**

दी गई आकृति में कितने त्रिभुज हैं?

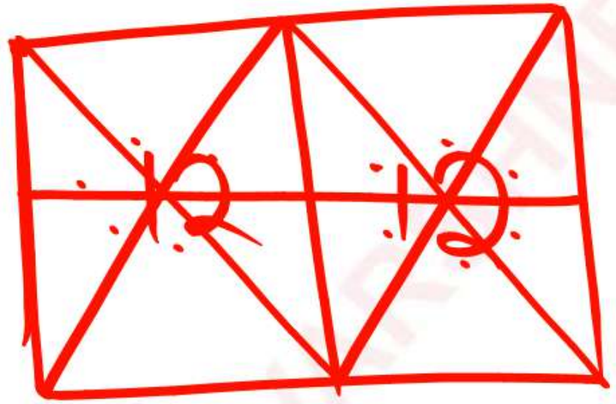


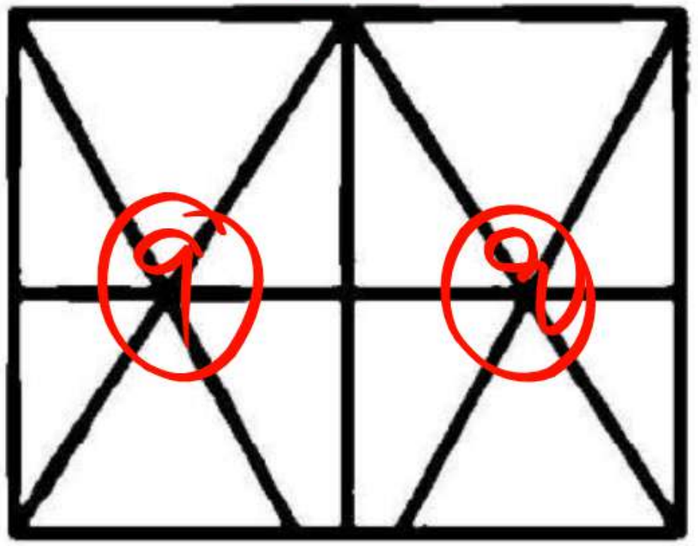
(a) 18
(b) 12
(c) 14
(d) 16

PIYUSH VARSHNEY SIR

46. **How many triangles are there in the given figure?**

दी गई आकृति में कितने त्रिभुज हैं?





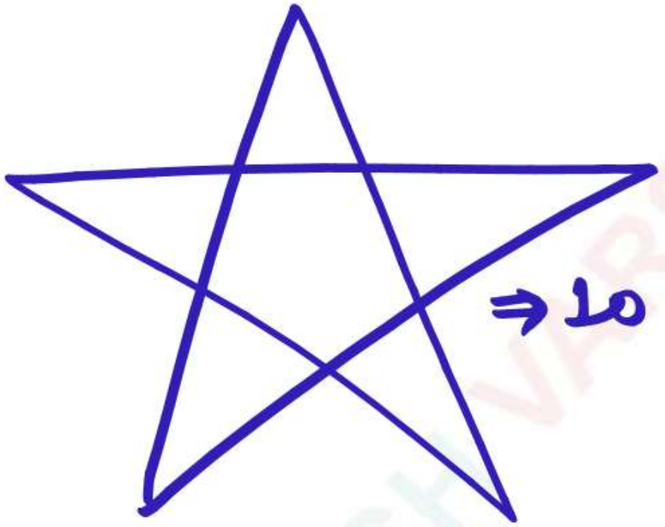
24 + 4 = 28

(a) 10 (b) 12

(c) 18 (d) 20

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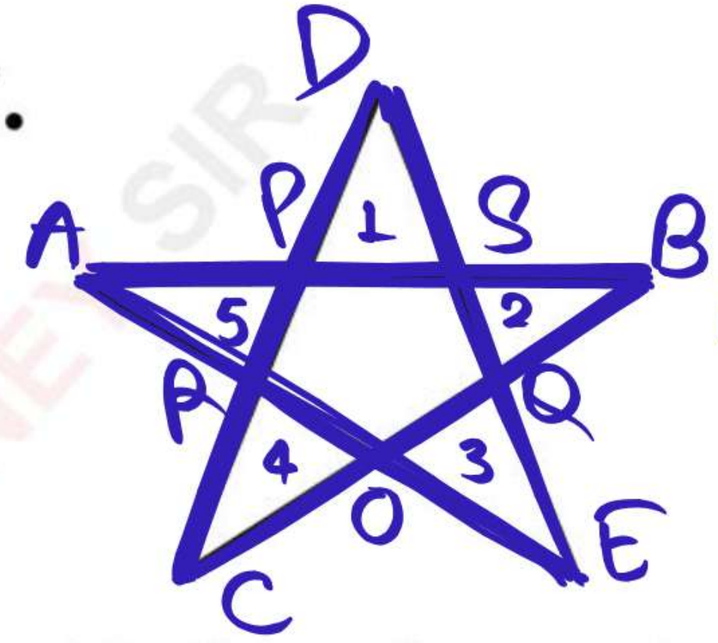
8



⇒ 10

5 = lines

47.



Basic Extra
5+ 5
= 10

ABO
BCP
CDQ
DER
EAS

Find out the number of triangles in the given figure?

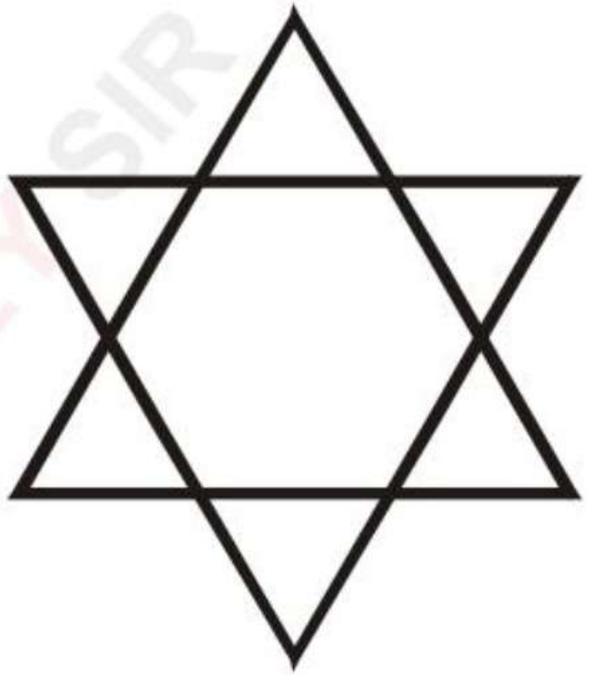
दी गई आकृति में त्रिभुजों की संख्या बताइए?

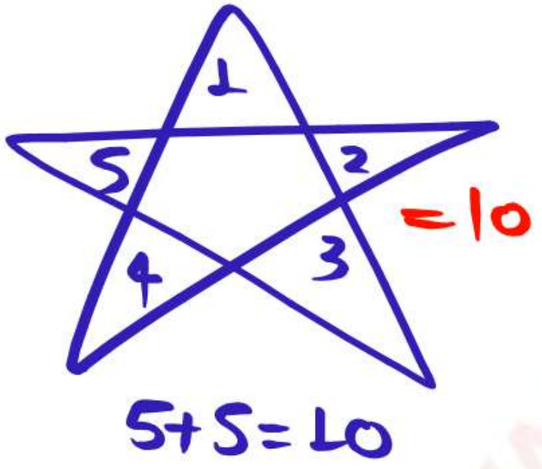
(a) 8 **(b) 7**

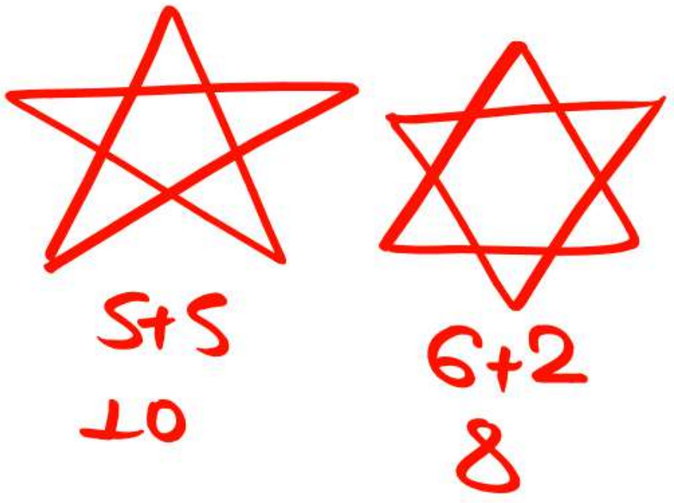
(c) 10 **(d) 12**

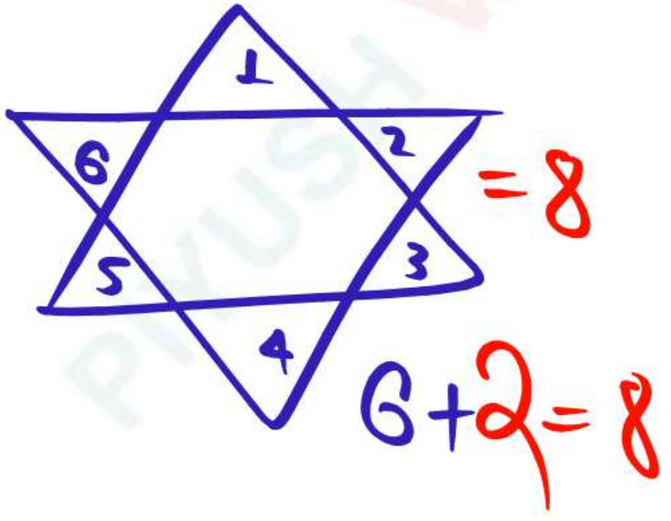
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48.









Find out the number of triangles in the given figure?

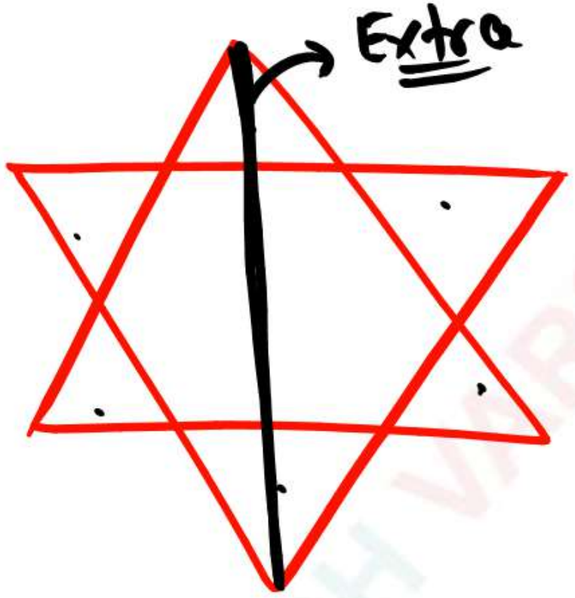
दी गई आकृति में त्रिभुजों की संख्या बताइए?

(a) 10 (b) 8

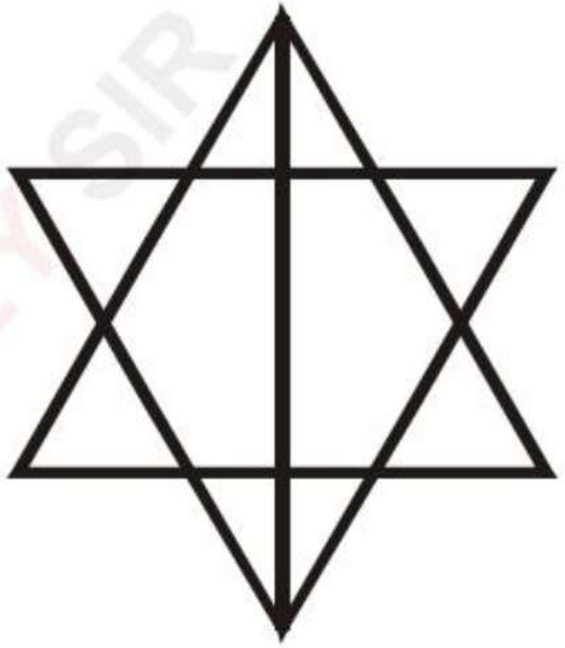
(c) 6 (d) 7

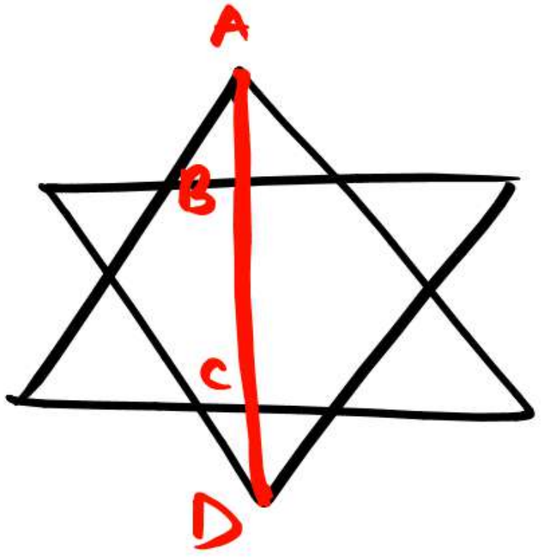
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49.



8+10





AB = 2.
AC = 2.
AD = 2.
DC = 2.
DB = 2.

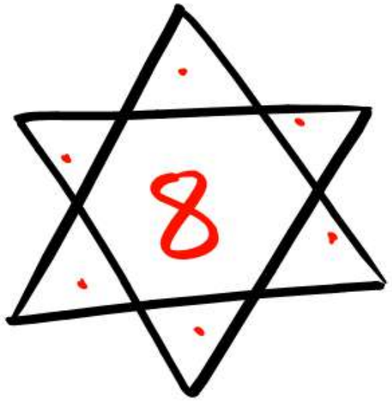
Find out the number of triangles in the given figure?

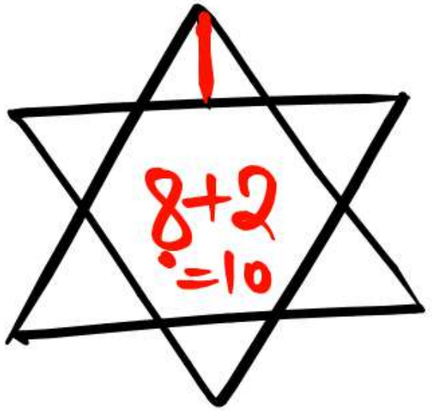
दी गई आकृति में त्रिभुजों की संख्या बताइए?

(a) 14 **(b) 20**

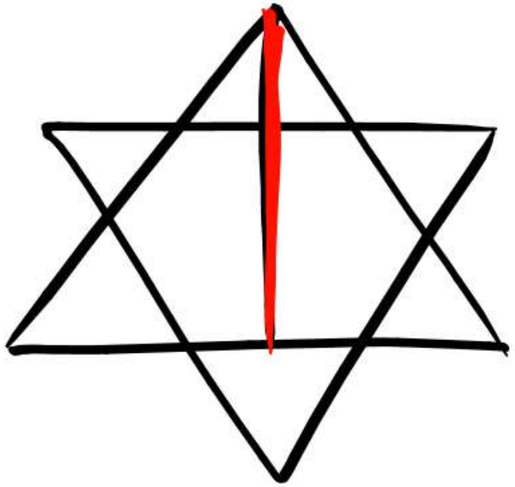
(c) 16 **(d) 18**

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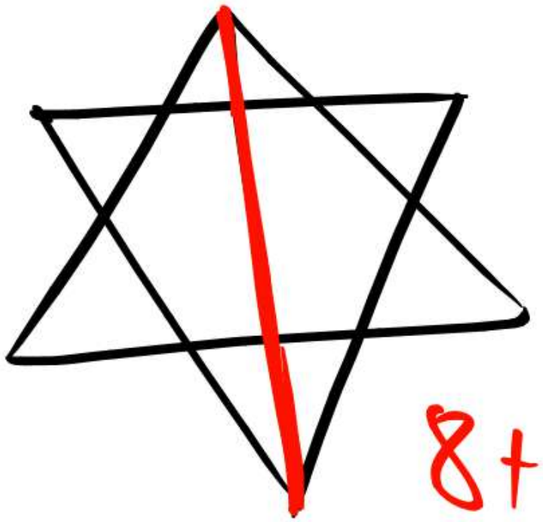




8+4
=12



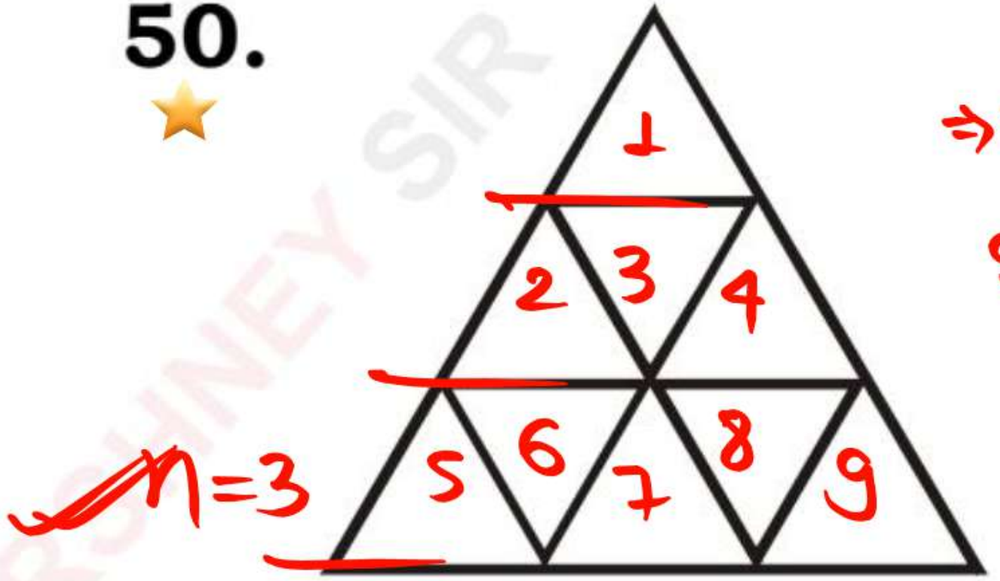




8+10=18

Type-3

50.



⇒ ×2-5

9×2-5=13

n=3

Find out the number of triangles in the given figure?

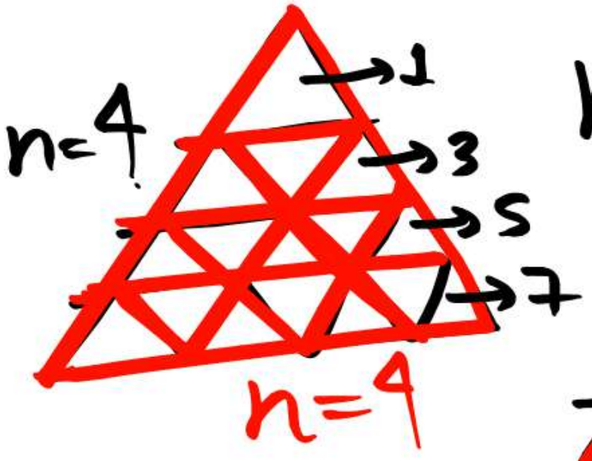
दी गई आकृति में त्रिभुजों की संख्या बताइए?

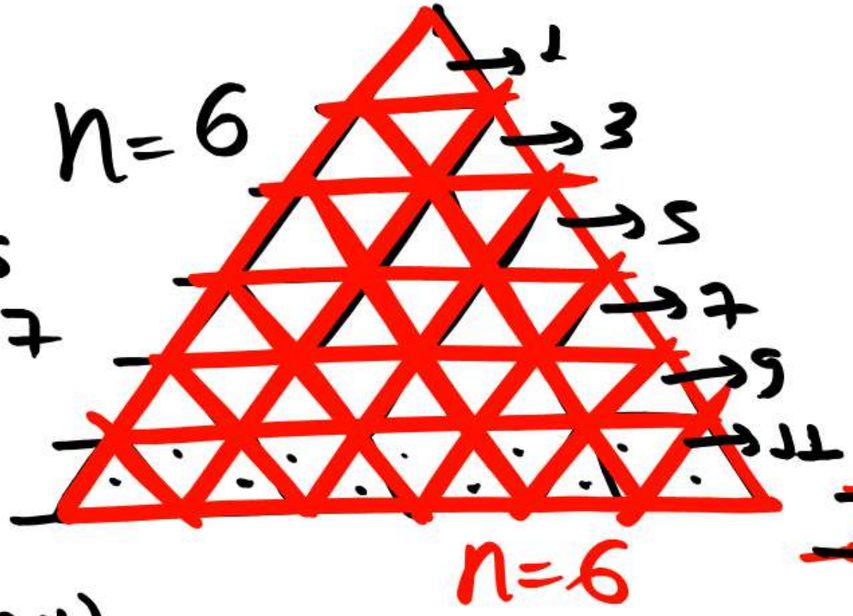
(a) 12 **(b) 13**

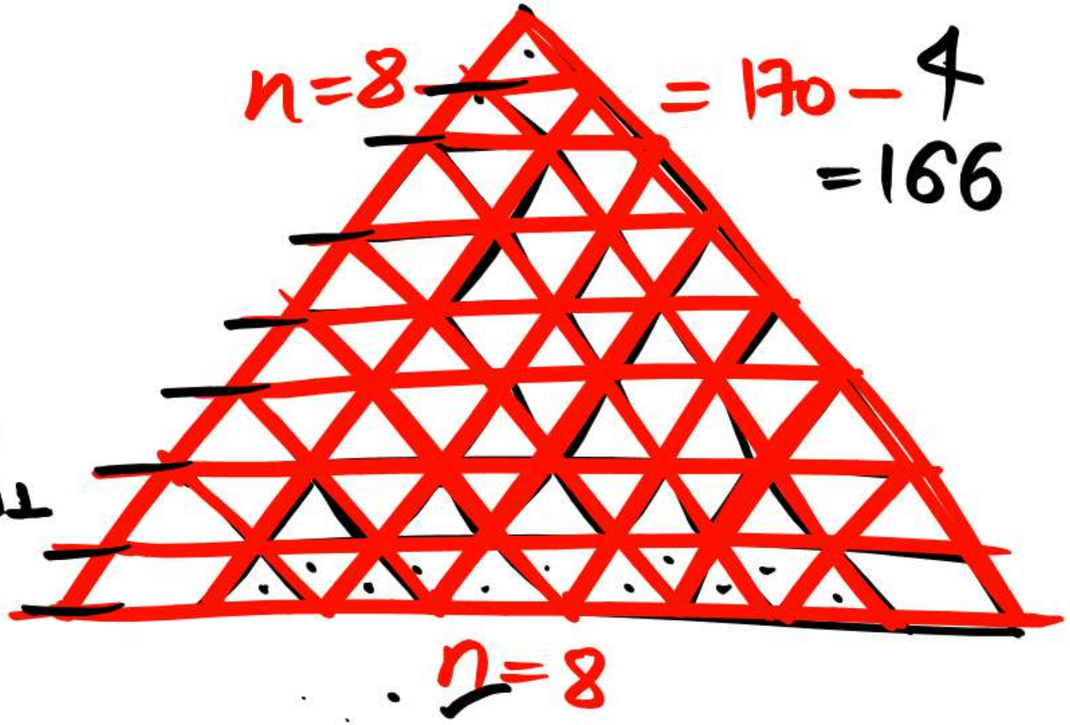
(c) 18 **(d) 20**

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n = even ✓ 2, 4, 6, 8, 10, ...







$$\Delta = \frac{n(n+2)(2n+1)}{8}$$

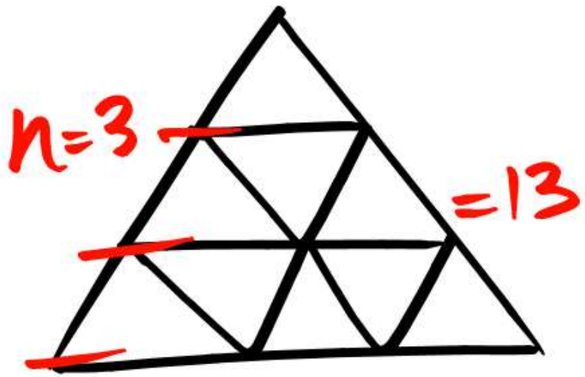
$$\frac{4(6)(9)}{8} = 27$$

$$\Delta = \frac{6 \times 8 \times 13}{8} = 78$$

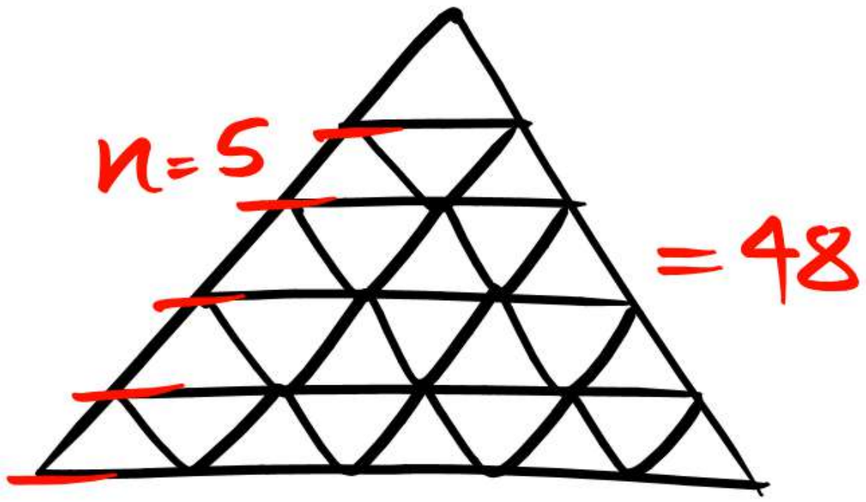
$$= 170 - 4 = 166$$

$$\Delta = \frac{8 \times 10 \times 17}{8}$$

$$= 170$$



=13



=48

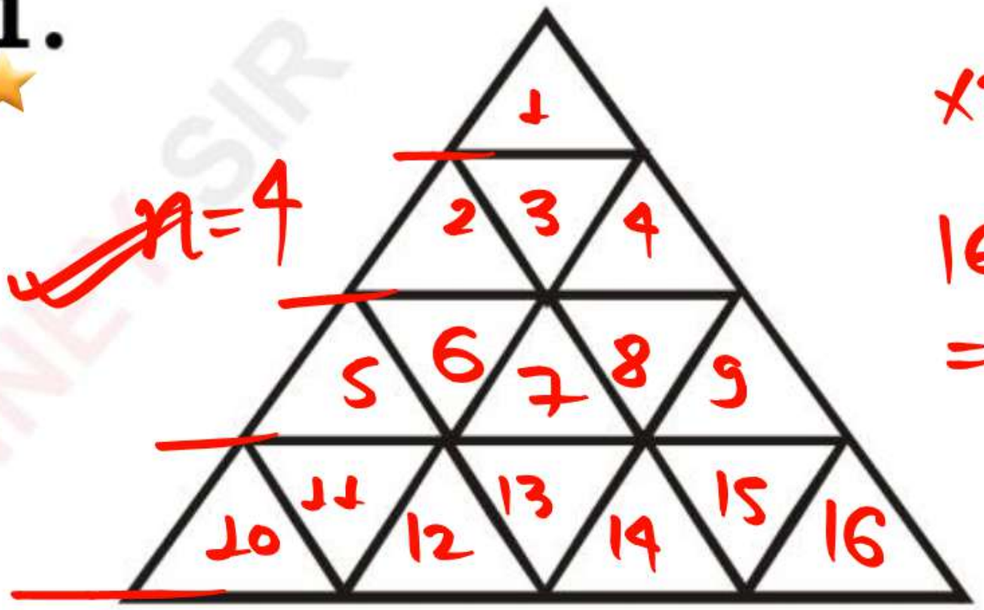
$$\Delta = \frac{[n(n+2)(2n+1)-1]}{8}$$

$$\frac{3(5)(7)-1}{8} = \frac{104}{8} = 13$$

$$\frac{5\times7\times11-1}{8} = \frac{384}{8}$$

$$= 48$$

51.



Find out the number of triangles in the given figure?

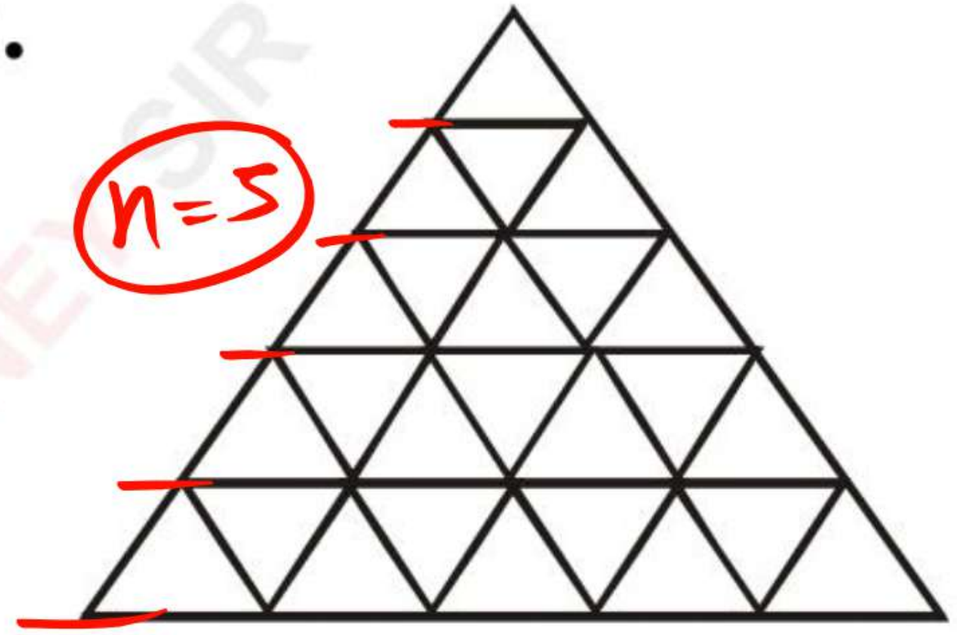
दी गई आकृति में त्रिभुजों की संख्या बताइए?

(a) 27
(b) 26
(c) 25
(d) 30

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Piyush Varshney Sir (Reasoning Expert)

52.



$\frac{5 \times 7 \times 11 - 1}{8}$

$= \frac{384}{8}$

$= 48$

Find out the number of triangles in the given figure?

दी गई आकृति में त्रिभुजों की संख्या बताइए?

(a) 41 **(b) 50**

(c) 43 **(d) 48**

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